

Scalable, Accurate, and Efficient Sampling Algorithms

For Generative Modeling and Bayesian Inference



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I dedicate this thesis to... ...

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Shreyas Padhy
April 2025

Acknowledgements

And I would like to acknowledge ...

Abstract

Sampling from high-dimensional and multimodal distributions is a fundamental challenge in modern machine learning applications, ranging from generative modelling to Bayesian inference. In this thesis, we propose a family of algorithms that address these challenges through a theoretical framework that enables highly accurate samples from arbitrary multimodal distributions, as well as scalable methods for sampling from very high-dimensional Gaussian distributions. First, we propose a stochastic calculus framework for designing scalable and accurate neural diffusion sampling algorithms. Additionally, we propose a unifying framework for the task of “conditional” generation, i.e., obtaining samples from probability distributions that satisfy additional constraints. We demonstrate the performance of these algorithms on tasks involving sampling from unnormalised distributions, as well as conditional generation for inverse problems and finetuning. Finally, we turn to the specific task of developing highly scalable Bayesian inference techniques by proposing algorithms to sample from high-dimensional Gaussian distributions, and novel algorithms to optimise the hyperparameters of Bayesian inference algorithms using samples. Through this lens, we unify inference and hyperparameter optimisation in Bayesian Neural Networks and Gaussian Processes, showing high-quality uncertainty estimates at Imagenet-scale.

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Symbols

Acronyms / Abbreviations

AIS Annealed Importance Sampling

CMCD Controlled Monte Carlo Diffusions

DDPM Denoising Diffusion Probabilistic Models

EBM Energy-Based Model

GAN Generative Adversarial Network

HMC Hamiltonian Monte Carlo

IS Importance Sampling

MCMC Markov Chain Monte Carlo

NUTS No U-Turn Sampler

RL Reinforcement Learning

SA Simulated Annealing

SDE Stochastic Differential Equation

ULA Unadjusted Langevin Annealing

VAE Variational Auto-Encoder

VI Variational Inference

Principal Notation

General

Ω	Sample space	26
$\mathcal{F}$	σ -algebra	26
$\mathbb{P}$	Probability measure	27
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space	27
$\mathcal{B}(\mathbb{R}^n)$	Borel σ -algebra	27
λ	Lebesgue measure	27
$\{X_t\}_{t \in T}$	Stochastic process	28
X_t	Stochastic process at time t	28

Chapter 1

Introduction

Machine learning has revolutionised the way we approach modelling and decision making in many fields, ranging from healthcare to scientific discovery. Modern algorithms are fundamentally defining a new approach to how we analyse data and derive insights from complex systems. While the entire gamut of machine learning applications and methodologies are vast and varied, there is often a common thread among them: the ability to obtain samples from complex, high-dimensional probability distributions. The ability to obtain samples from these complex distributions has allowed us to develop powerful applications, such as image synthesis, protein structure prediction, and drug design, to name a few. A large family of methods focus on the task of *generative modelling*, which is the task of estimating a probability distribution when provided with samples from it, in order to obtain new, novel samples. The approaches have evolved from one-shot sampling approaches such as GANs to iterative samplers like diffusion models, dramatically improving the quality of samples at the expense of speed. Another family of ML algorithms focus on the task of *obtaining samples from unnormalised probability distributions* where we can only evaluate the density of a given sample. Here, classical MCMC techniques remain indispensable, though there have been recent innovations in neural sampling algorithms that boost efficiency and accuracy.

The ability to estimate complex probability distributions and draw samples has the potential to unlock really impactful downstream applications as well. Many modern *Bayesian inference* algorithms rely on approximating quantities of interest using high-quality samples from the posterior distribution. While MCMC methods like HMC and NUTS provide this quality at an exorbitant cost, variational and Gaussian

approximations that enable efficient sampling are often quite competitive. The field of *Reinforcement Learning (RL)* also makes heavy use of sampling algorithms for learning policies and exploring the state-action space. Even in the modern *Large Language Model (LLM)* era, it has been shown that *how we sample* matters enormously: strategies like nucleus and temperature sampling have boosted the generation of coherent yet diverse, interesting text.

Across these domains and applications, we see a common theme. High dimensionality is a difficult challenge to overcome: techniques that leverage gradients (Langevin, HMC) or adaptive truncation (nucleus sampling) can help tame the curse of dimensionality by guiding samples towards high-density regions. Secondly, there is always a trade-off between exactness and efficiency: e.g. exact MCMC vs faster VI in Bayesian inference algorithms, or exhaustive beam search vs stochastic sampling in text generation. Algorithms that can improve along these two axes have immense value in both allowing us to scale these techniques over larger and more complex data, as well as improve the accuracy of existing methods. While these fields are often explored independently in the literature, there has historically been cross-pollination between fields: ideas from statistical physics such as annealing and tempering now form the basis of the best sampling techniques; probabilistic inference tools such as HMC and VI have been adapted and scaled for deep learning; and better deep learning models contribute back to better neural samplers and generative models. As models and data grow in size, and our desire to apply machine learning to ever more ambitious domains grows, efficient and accurate sampling techniques will remain a crucial area of research.

Furthermore, there have been many threads of promising research in the sampling literature, that approach the problem from different viewpoints. Denoising diffusion probabilistic models (DDPM) designed discrete Markov chains that gradually added noise to data and then learnt to reverse this noising process using variational inference. Score matching techniques were proposed independently that converted the task of learning probability distributions into learning the gradients of their logarithms, a quantity termed the “score”. Simultaneously, there has been a resurgence in normalising flows and flow matching techniques, where algorithms attempt to learn continuous normalising flows to map different probability distributions to each other. In the statistical inference literature, methods involving Langevin dynamics have been quite popular, which is a classic MCMC technique that follows continuous-time Markov processes where the probability distribution of interest is the stationary distribution. This field is ever growing, and there are advancements across variational inference,

optimal transport theory, and stochastic control, to name a few, that are often hard to view under a unifying lens. This is a problem that we hope to remedy in this thesis, by developing a theoretical framework that can recast ideas from disparate fields into a unified viewpoint.

1.1 Thesis outline and contributions

The goal of this thesis is twofold: we develop a strong underlying mathematical framework of stochastic calculus to unify sampling, generative modelling and conditional sampling algorithms under the umbrella of stochastic differential equations, and we propose a sampling-based viewpoint of Bayesian inference, allowing us to scale algorithms to massive datasets and models. The thesis is organised as follows.

- Chapter 2 introduces the sampling problem, and proposes a heirarchical variational inference framework that would allow us to unify many generative modelling and sampling algorithms under a single framework of loss functions and optimisation. Along the way, we discuss popular algorithms such as Annealed Importance Sampling, Score Matching, and Diffusion Models, and show how they converge to stochastic differential equations in the continuous limit. This connection motivates us to study the variational framework under a stochastic calculus framework, and eventually allow us to develop novel algorithms due to this theoretical framework.
- Chapter 3 introduces stochastic differential equations, along with the theory of Ito calculus, which allows us to develop “path”-based rules of probability theory that will be useful in this thesis. Further, we show two specific ways of time-reversal in SDEs, and discuss how different theoretical results can be derived from these two different perspectives and their drawbacks. Finally, we show how it is difficult to write probabilistic divergences under a “reverse-time” framework under regular Ito calculus, and therefore introduce an individual path-based framework of Follmer’s Ito calculus.
- In Chapter 4, we use our stochastic calculus framework to develop that Controlled Monte Carlo Diffusion (CMCD) sampler, an algorithm that connects heirarchical variational inference with diffusion models to adapt both forward and backward dynamics within diffusion processes, yielding an approach that has strong theoretical guarantees while maintaining remarkable practical efficiency. We then explore the related task of conditional sampling, i.e. the task of modifying sampling

algorithms to target specific conditions, and unify existing approaches under a stochastic calculus framework through Doob’s h -transform.

- Chapter 4 introduces a somewhat non-traditional viewpoint of Bayesian inference that approximates many components of the Bayesian methodology using a finite number of samples from the posterior distribution. Recognising that many practical Bayesian inference tasks do not necessarily require sampling from the full, exact posterior but instead can leverage high-dimensional Gaussian approximations, we dedicate this part of the thesis to scalable Gaussian sampling methodologies. We first show how Gaussian processes and Bayesian neural networks can be unified through the linearised Laplace approximation, allowing us to simplify the task of Bayesian inference to that of sampling from high-dimensional Gaussians. We then demonstrate the surprising empirical performance these linearised approximations, which despite their simplicity, offer highly competitive uncertainty quantification performance.
- Chapter 5 is dedicated to the development of a family of Bayesian inference algorithms that can efficiently draw samples from very high-dimensional Gaussian distributions using stochastic optimisation techniques, and use these samples to tune model hyperparameters, providing a highly scalable approach to model selection. We develop theoretical bounds on the quality of model selection using a finite number of samples, as well as practical insights and suggestions for speeding up stochastic optimisation. Our algorithms demonstrate significant scalability improvements, successfully enabling Bayesian inference on large-scale neural networks such as ResNet-50 models on an Imagenet scale, as well as the ability to train and perform inference with Gaussian processes on millions of datapoints without any approximations.

1.2 List of publications in the thesis

Here, I list the papers that form a part of this thesis.

1. “Transport meets Variational Inference: Controlled Monte Carlo Diffusions”. F. Vargas*, **S. Padhy***, D. Blessing, N. Nüsken. In *International Conference on Learning Representations (ICLR)*, 2024.
2. “DEFT: Efficient Finetuning of Conditional Diffusion Models by Learning the Generalised h -transform”. A. Denker*, F. Vargas*, **S. Padhy***, K. Didi*, S.

- Mathis*, V. Dutordoir, R. Barbano, E. Mathieu, U. Komorowska, P. Liò. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
3. “Sampling-based inference for large linear models, with application to linearised Laplace”. J. Antorán*, **S. Padhy***, R. Barbano, E. Nalisnick, D. Janz, J.M. Hernández-Lobato. In *International Conference on Learning Representations (ICLR)*, 2023.
 4. “Sampling from Gaussian process posteriors using stochastic gradient descent”. J.A. Lin*, J. Antorán*, **S. Padhy***, D. Janz, J.M. Hernández-Lobato, A. Terenin. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
 5. “Stochastic Gradient Descent for Gaussian Processes Done Right”. J.A. Lin*, **S. Padhy***, J. Antorán*, A. Tripp, A. Terenin, C. Szepesvári, J.M. Hernández-Lobato, D. Janz. In *International Conference on Learning Representations (ICLR)*, 2024.
 6. “Improving Linear System Solvers for Hyperparameter Optimisation in Iterative Gaussian Processes”. J.A. Lin, **S. Padhy**, B. Mlodozieniec, J. Antorán, J.M. Hernández-Lobato. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
 7. “Iterative Importance Fine-tuning of Diffusion Models”. A. Denker, **S. Padhy**, F. Vargas, J. Hertrich. In *Frontiers in Probabilistic Inference workshop at International Conference on Learning Representations (ICLR)*, 2025.
 8. “No Trick, No Treat: Pursuits and Challenges Towards Simulation-free Training of Neural Samplers”. J. He, Y. Du, F. Vargas, D. Zhang, **S. Padhy**, R. OuYang, C. Gomes, J.M. Hernández-Lobato. In *Frontiers in Probabilistic Inference workshop at International Conference on Learning Representations (ICLR)*, 2025.

Chapter 2

An Introduction to the Sampling Problem

The task of sampling in machine learning can be broadly divided into three distinct paradigms. In the first situation, we don't have any existing samples from an unnormalised probability distribution, and we are only able to evaluate the unnormalised density $\hat{\pi}(x)$ and the gradient of the density $\nabla_x \hat{\pi}(x)$ ¹. The task of *unnormalised density sampling* (Figure 2.1, I.) focuses on learning to draw samples from this probability distribution (and often-times using these samples to estimate the normalising constant).

In the second task of *generative modelling* (Figure 2.1, II.), we only have access to a training dataset of samples from our probability distribution of interest, i.e. $\mathcal{D}_{\text{train}} = \{x^{(i)}\}_{i=1}^{N_{\text{train}}}$, but not the ability to evaluate the density or its gradients. The task therefore focuses on using these samples to learn parametrised approximations to our probability distribution of interest, allowing us to draw novel samples from this approximation, and sometimes allowing us to estimate the density $\hat{\pi}(x)$.

Finally, in the third task of *conditional sampling*, we consider two distinct subcategories. In the first situation, we already have access to a parametrised approximation to the distribution of interest $\hat{\pi}$, and we wish to learn a parametrised approximation to a conditional distribution $\pi_{\cdot|y}$ that is specified through the definition of a likelihood term $p(y | x)$, for some paired observations y . We call this the task of *conditional generative modeling* (Figure 2.1, III.A). In the second situation, we have access to samples from the unconditional distribution $\hat{\pi}$, and we wish to learn how to modify

¹The assumption is often made that one can evaluate the gradient of the log-density $\nabla_x \log \hat{\pi}$. These are equivalent assumptions, as by the chain rule of calculus we can write $\nabla_x \log \hat{\pi}(x) = \frac{1}{\hat{\pi}(x)} \nabla_x \hat{\pi}(x)$

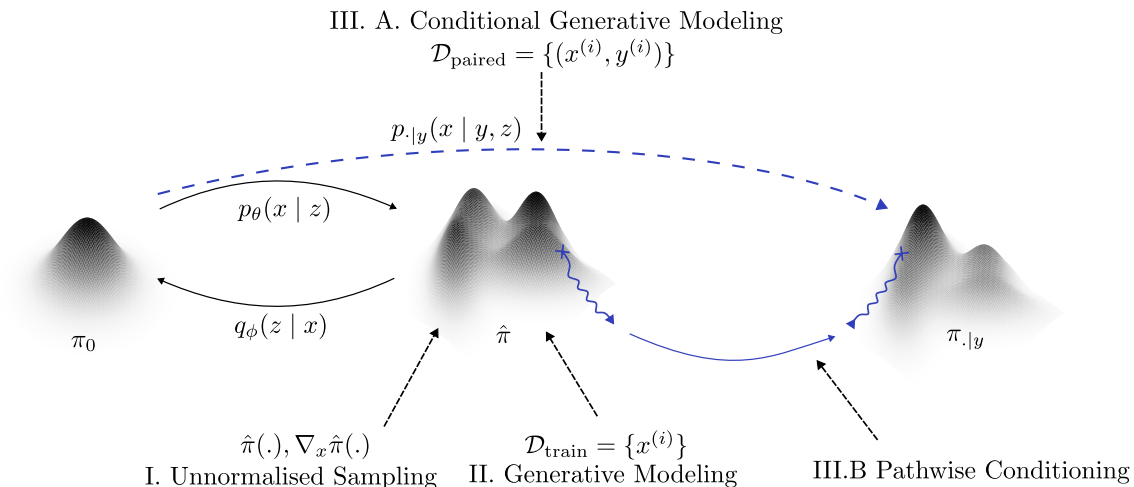


Fig. 2.1. The sampling problem can be broadly split into three paradigms: (I) Unnormalised density sampling, where we are only able to evaluate the unnormalised density of a sample $\hat{\pi}$, and must learn how to draw samples from this probability distribution. (II) Generative modelling, where we only have access to a training dataset of samples from our probability distribution of interest, i.e. $\mathcal{D}_{\text{train}} = \{x^{(i)}\}_{i=1}^{N_{\text{train}}}$, but not the ability to evaluate the density or its gradients. (III.A) Conditional generative modelling, where we already have access to a parametrised approximation to $\hat{\pi}$, and wish to learn an approximation to the conditional distribution $\pi_{\cdot|y}$ using an additional paired dataset $\mathcal{D}_{\text{paired}}$. (III.B) A special case of conditional sampling, where we have access to samples from the distribution of interest $\hat{\pi}$ (usually a simple family of distributions such as Gaussians), and we wish to modify these samples so that would belong to the conditional distribution $\pi_{\cdot|y}$.

these samples to instead belong to the conditional distribution $\pi_{\cdot|y}$. We call this the task of “*pathwise*” conditioning (Figure 2.1, III.B), where the term “path” refers to the fact that we are modifying individual samples, rather than the parametrised approximation itself. Usually, the “pathwise” conditioning setup is only possible when the unconditional distribution $\hat{\pi}$ as well as the likelihood term $p(y | x)$ belong to a simple family of distributions, such as Gaussian distributions. In the task of Bayesian inference, practitioners may recognise conditional sampling as the task of posterior inference, where the posterior distribution $\pi_{\cdot|y} \propto p(y | x)\hat{\pi}(x)$.

2.1 A Unifying Variational Framework

In this thesis, we will focus on a framework for designing sampling algorithms that has found great success across a variety of algorithms, whether implicitly or as an explicit design. In simple terms, we design (or learn) a mapping between a simple base

distribution μ and the distribution of interest $\hat{\pi}$. By design, it is easy to obtain samples $z \sim \mu$, and we attempt to learn a mapping $p_\theta(x | z)$ that transports these samples to samples from the distribution of interest $\hat{\pi}$. In many algorithms, it is actually simpler to first design a “noising”, i.e. a reverse mapping $q_\phi(z | x)$ that takes samples $x \sim \hat{\pi}$ and maps them to the simpler distribution μ . Under this framework, the choice of μ and the mappings p_θ and q_ϕ cover a large family of algorithms, allowing for an infinite number of valid mappings to exist. In order to constrain our solution space to tractable algorithms, we usually will make a choice for μ that is easy to sample from, and the mappings p_θ and q_ϕ are chosen to be parametrised functions that are easy to manipulate and compute. In this setup, it is then natural to consider a loss function of the following form to learn the mappings

$$\mathcal{L}_D(\theta, \phi) = \mathcal{D}(q_\phi(z | x)\mu(z) || p_\theta(x | z)\hat{\pi}(x)). \quad (2.1)$$

For this thesis, we make a specific choice of a heirarchical latent variable model for our mappings p_θ and q_ϕ . While this choice might seem arbitrary, as we see later, a large family of existing algorithms fall under this framework, including Annealed Importance Sampling (AIS) algorithms (Neal, 2001; Jarzynski, 1997a; Jarzynski, 1997b), Score Matching algorithms (Song and Ermon, 2019; Song, Sohl-Dickstein, et al., 2020), and Denoising Diffusion Probabilistic Models (DDPM) (Ho et al., 2020).

We make the choice of $x_0 := z$ and $x_T := x$ with a series of intermediate latent variables $\{x_1, \dots, x_{T-1}\}$. The mappings p_θ and q_ϕ are then chosen to be simple and easy to compute in a heirarchical manner as

$$p_\theta(x_T, \dots, x_1 | x_0) = \prod_{t=0}^{T-1} p_{\theta_t}(x_{t+1} | x_t), \quad q_\phi(x_0, \dots, x_{T-1} | x_T) = \prod_{t=0}^{T-1} q_{\phi_t}(x_t | x_{t+1}). \quad (2.2)$$

While we can choose arbitrarily complex functions to define the intermediate mappings p_{θ_t} and q_{ϕ_t} , as $T \rightarrow \infty$ and we consider the continuous limit, even a simple choice for p_{θ_t} and q_{ϕ_t} such as Gaussian distributions can yield highly flexible mappings. Furthermore, making a Gaussian choice for p_{θ_t} and q_{ϕ_t} allows us to write down the continuous-time limit of the mappings as SDEs, which will enable us to use the rich mathematical framework of stochastic calculus to develop more flexible algorithms later. The Gaussian choice for p_{θ_t} and q_{ϕ_t} is given by

$$p_{\theta_t}(x_{t+1} | x_t) = \mathcal{N}(x_{t+1}; x_t + f_{\theta_t}(x_t), \sigma_t^2 I), \quad q_{\phi_t}(x_t | x_{t+1}) = \mathcal{N}(x_t; x_{t+1} + b_{\phi_t}(x_{t+1}), \sigma_t^2 I), \quad (2.3)$$

for some choice of time-varying functions f , b , and σ . Writing down the divergence between the extended latent variable mappings, and using the data-processing inequality (Cover, 1999), we can show that the divergence between the extended mappings is an upper bound over Equation (2.1),

$$\mathcal{D}_{KL}(q_\phi(x_0, \dots, x_T \parallel p_\theta(x_0, \dots, x_T)) \geq \mathcal{D}(q_\phi(z \mid x)\mu(z) \parallel p_\theta(x \mid z)\hat{\pi}(x)). \quad (2.4)$$

As an aside, we note that we can write down the heirarchical mappings in the form of Markov chains with forward and backward transition densities defined through the forward and backward transition kernels F_{θ_t} and B_{ϕ_t} as

$$x_{t+1} \leftarrow x_t + f_{\theta_t}(x_t) + \sigma_t \epsilon_{t+1}, \quad F_{\theta_t}(x_{t+1} \mid x_t) = \mathcal{N}(x_{t+1}; x_t + f_{\theta_t}(x_t), \sigma_t^2 I), \quad (2.5)$$

$$x_t \leftarrow x_{t+1} + b_{\phi_t}(x_{t+1}) + \sigma_t \epsilon_t, \quad B_{\phi_t}(x_t \mid x_{t+1}) = \mathcal{N}(x_t; x_{t+1} + b_{\phi_t}(x_{t+1}), \sigma_t^2 I), \quad (2.6)$$

where $\{\epsilon_t\}_{t=1}^T$ are i.i.d. standard normal random variables.

Remark: The Continuous Limit

In the limit of infinite steps, where $T \rightarrow \infty$, the transition densities governing the mappings between latent variables t and $t + 1$ can be written down as the solutions to a pair of stochastic differential equations (SDEs)^a given by

$$dX_t = f_{\theta_t}(X_t)dt + \sigma_t dW_t, \quad dX_t = b_{\phi_t}(X_t)dt + \sigma_t d\overleftarrow{W}_t, \quad (2.7)$$

where dW_t is the “forward” Wiener process, and $d\overleftarrow{W}_t$ is the “backward” Wiener process. In this continuous limit, the joint distribution of the latent variables further converge to a quantity termed the “path” measure of the corresponding SDEs, which we refer to as $\overrightarrow{\mathbb{P}}_\theta$ and $\overleftarrow{\mathbb{P}}_\phi$ respectively. If we are able to define a valid probability divergence measure between the forward and backwards path measure, we can rewrite Equation (2.1) in the continuous limit as

$$\mathcal{L}_D(\theta, \phi) = \mathcal{L}_D(\overrightarrow{\mathbb{P}}_\theta \parallel \overleftarrow{\mathbb{P}}_\phi). \quad (2.8)$$

As we will motivate in Chapter 3, it is non-trivial to easily define a valid probability divergence measure between the forward and backwards path measure, and as one of our main contributions, we will develop a unique change of measure between forward and backwards path measures that will enable us to use this framework to represent a large family of existing algorithms.

^aThis remark is a brief introduction to the continuous limit of the mappings p_{θ_t} and q_{ϕ_t} . We introduce the notation and terminology more rigorously in Chapter 3.

Let us now discuss the individual paradigms of the sampling problems, as well as a few existing algorithms that have found great success and adoption in the machine learning community. Specifically, we choose to focus on algorithms that are readily amenable to the framework we have introduced, which is not necessarily a big restriction, as the algorithms we choose to focus on are some of the most popular algorithms in the field.

2.2 Unnormalised Density Sampling

When a probability distribution π is known only up to a normalisation constant, drawing samples from it is often non-trivial. Such distributions arise quite regularly in Boltzmann distributions such as in statistical physics and graphical models, and in modern energy-based models (EBMs). This issue is also quite prevalent in Bayesian inference applications where sampling from the posterior distribution is intractable due to an inability to estimate the normalising constant. To make our notation concrete, we write down π in terms of the unnormalised probability density $\hat{\pi}$ and the normalising constant Z as

$$\pi = \frac{\hat{\pi}}{Z}, \quad Z = \int \hat{\pi}(x) dx. \quad (2.9)$$

A related problem is that of *partition function estimation*, which is a special case of the problem of *estimating expectations of functions* under the distribution π . We can write down the expectation of a function f under π as

$$\mathbb{E}_{\pi} [f(x)] = \frac{\int f(x) \hat{\pi}(x) dx}{\int \hat{\pi}(x) dx} = \frac{\mathbb{E}_{\hat{\pi}} [f(x)]}{Z}. \quad (2.10)$$

and correspondingly, the partition function Z can be written as

$$Z = \int \hat{\pi}(x) dx = \mathbb{E}_{\hat{\pi}} [\mathbb{I}]. \quad (2.11)$$

2.2.1 Importance Sampling

Importance Sampling (IS) algorithms tackle the problem of estimating expectations under π by using a simpler approximating distribution q that is easy to sample from.

By a simple manipulation of the expectation, we can write

$$\mathbb{E}_\pi[f] = \frac{1}{Z} \int f(x) \hat{\pi}(x) dx \quad (2.12)$$

$$= \frac{1}{Z} \int f(x) \frac{\hat{\pi}(x)}{q(x)} q(x) dx \quad (2.13)$$

$$= \frac{1}{Z} \mathbb{E}_q \left[f(x) \underbrace{\frac{\hat{\pi}(x)}{q(x)}}_{w(x)} \right] \quad (2.14)$$

Further, from Equation (2.11), we can see that the partition function Z is simply the expectation of the indicator function under π , and so we can estimate Z by drawing samples from q and computing the empirical average of $w(x)$ over these samples, giving us

$$\hat{Z} = \frac{1}{N} \sum_{i=1}^N w(x^{(i)}), \quad x^{(i)} \sim q \quad (2.15)$$

$$\mathbb{E}_\pi[f] \simeq \frac{1}{\hat{Z}} \frac{1}{N} \sum_{i=1}^N f(x^{(i)}) w(x^{(i)}) \quad (2.16)$$

Unfortunately, there are a few issues with this approach in general. Firstly, the choice of q is critical to the performance of the algorithm, and in general, it is difficult to choose a q that is both close to π and easy to sample from. Secondly, IS techniques don't provide us with a way to design an approximation to π that we can draw additional samples from, and we can merely use samples from q to estimate expectations under π . We therefore turn our attention to Markov Chain Monte Carlo (MCMC) algorithms, which can be used to sample from π directly, and later, we see how we can consider an Importance Sampling viewpoint of MCMC algorithms that allow us to accurately estimate expectations under π .

2.2.2 MCMC Algorithms and Langevin Dynamics

Markov Chain Monte Carlo (MCMC) methods allow us to construct a Markov chain with an equilibrium distribution that matches the target distribution π . While the literature of MCMC algorithms is vast, and we could dedicate an entire chapter to summarising the recent advances, for the purpose of this thesis, we will focus on one specific algorithm: Langevin dynamics, which can exploit information about the gradient of the log-density of $\hat{\pi}$ to obtain high-quality samples compared to other MCMC variants. Specifically, assuming a sample from a simple distribution $x_0 \sim \mu$,

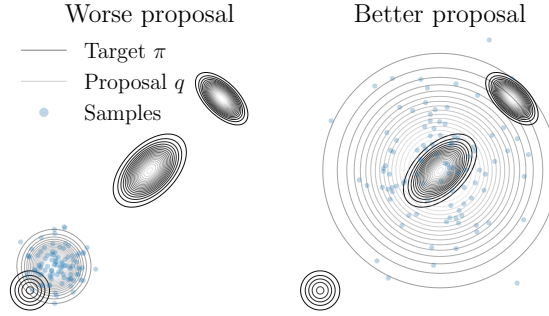


Fig. 2.2. The target density $\pi(x)$ is the three-component Gaussian mixture with weights $w = (0.6, 0.3, 0.1)$, means $\mu_1 = (0, 0)$, $\mu_2 = (1.5, 1.5)$, $\mu_3 = (-2.5, -2.5)$ and covariances $\Sigma_1 = \begin{bmatrix} 0.10 & 0.05 \\ 0.05 & 0.10 \end{bmatrix}$, $\Sigma_2 = \begin{bmatrix} 0.05 & -0.03 \\ -0.03 & 0.05 \end{bmatrix}$, $\Sigma_3 = 0.05 I$. We compare the coverage provided by two choices of importance sampling distribution q , given by $q_1(x) = \mathcal{N}([-2, -2]^\top, 0.1 I)$ and $q_2(x) = \mathcal{N}(0, I)$. 100 random samples (blue dots) illustrate their overlap with π , showing that the choice of importance sampling distribution can have a large impact on the performance of the algorithm.

we can follow the following iterative procedure

$$x_{t+1} \leftarrow x_t + \Delta t \nabla_x \log \hat{\pi}(x) + \sqrt{2\Delta t} \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I), \quad t = 0, \dots, T \quad (2.17)$$

where Δt is a small interval. When $\Delta t \rightarrow 0$ and $T \rightarrow \infty$, it can be shown that x_T obtained by the procedure above converges to a sample from $\pi(x)$ under some regularity conditions.

Vanilla Langevin dynamics can struggle in practise to produce samples that are diverse over the entire support of $\hat{\pi}$, as we use the gradient of the log-density of $\hat{\pi}$ to iteratively update the sample, and this can result in the samples being trapped in local modes of the density. We see next approaches that tackle this issue.

Remark: Continuous Limit of Langevin Dynamics

In fact, our choice of using t to refer to the discrete index foreshadows what Langevin dynamics converges to in the continuous limit: a time-varying stochastic differential equation. Intuitively, we could write down a differential equation of x w.r.t t as

$$\frac{dx_t}{dt} = \nabla_x \log \hat{\pi}(x_t) + C \times \frac{d\text{"noise"}}{dt}, \quad (2.18)$$

for some constant C . The introduction of a noise term in the differential equation introduces subtle issues that prevent us from relying on traditional ordinary

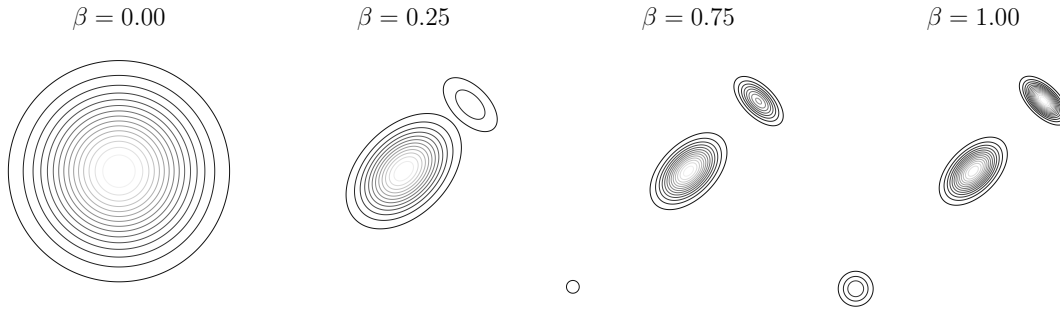


Fig. 2.3. Geometric annealing defines $\pi_\beta(x) = \mu(x)^{1-\beta} \pi(x)^\beta$ using the base proposal $\mu(x) = \mathcal{N}(0, I)$. Panels show π_β for $\beta = 0, 0.25, 0.75, 1$. The target density $\pi(x)$ is the same three-component Gaussian mixture as in Figure 2.2.

differential equation machinery, and we will need to develop a “stochastic” calculus framework in order to successfully find solutions for equations of such form. However, here, it suffices to say that Langevin dynamics is a specific instantiation of an SDE.

2.2.3 Simulated Annealing

Simulated Annealing (SA) (Kirkpatrick et al., 1983) was introduced as a way to improve the mixing/coverage of MCMC sampling. Instead of designing a Markov Chain that directly samples from $\pi = \pi_T$, we design a Markov Chain that samples from a sequence of distributions $\pi_0, \pi_1, \dots, \pi_T$ that interpolate between a simple distribution π_0 and the target distribution π_T . We first run a Markov Chain designed to sample from π_0 for a number of steps (usually less than required for equilibrium). Using these samples at a starting point, we then simulate a Markov Chain for π_1 , and so on. π_0 is usually designed to have really good coverage over the space where different modes for π might exist, allowing for good mixing.

While we can obtain samples covering multiple modes using SA, there are unfortunately no guarantees on obtaining correct results (i.e. each mode of π is represented with the correct probability), or estimating partition functions and expectations in a variance-reduced way. In order to solve these issues, we turn to another method that views this procedure as an importance sampling problem.

2.2.4 Annealed Importance Sampling

Annealed Importance Sampling (AIS) (Neal, 2001; Jarzynski, 1997a; Jarzynski, 1997b) was proposed independently by Radford Neal for statistical applications, as well as by Christopher Jarzynski for estimating free energies. Effectively, Radford Neal provided an importance sampling viewpoint of annealing, which allows us to assign weights to the samples obtained from different annealing runs, and use these weights to converge to the correct distribution as the number of runs increases. As before, AIS introduces a series of intermediate distributions starting from a simple distribution $\pi_0 := \mu$ until the distribution of interest $\pi_T := \pi$.

We first define a sequence of unnormalised distributions $\hat{\pi}_t(x)$ that are proportional to the annealed distributions $\pi_t(x)$, with $\hat{\pi}_T(x) = \hat{\pi}(x)$. Second, for each intermediate distribution $\hat{\pi}_t$, we require that we can simulate a Markov Chain using a forward transition density F_t that has π_t as its equilibrium distribution. Given these two design choices (of $\hat{\pi}_t$ and F_t), the algorithm proceeds by first sampling $x_0 \sim \pi_0$, then simulating a Markov chain using F_t for $t = 1$ with x_0 as the starting point to obtain x_1 , and so on until T . Usually, we run the chain at each step t for a fixed number of steps, and then use the final sample x_T as the sample from π .

In order to calculate the importance sampling weights for the T -th distribution, we would need to calculate $w_{\text{mar}}(x_T) = \hat{\pi}(x_T)/q_T(x_T)$. In order to do this, we first consider the extended space defined over the path that AIS takes, i.e. jointly over the points $(x_0, x_1, \dots, x_T)$. The effective proposal distribution over this extended space is given by

$$Q(x_{0:T}) = \pi_0(x_0) \prod_{t=1}^T F_t(x_t | x_{t-1}). \quad (2.19)$$

Therefore, the marginal proposal distributions at any intermediate step q_t would be defined as

$$q_t(x_t) = \int q_{t-1}(x_{t-1}) F_t(x_t | x_{t-1}) dx_{t-1} \text{ for } t \geq 1, \quad q_0(x_0) = \pi_0(x_0). \quad (2.20)$$

Unfortunately, we see that this quantity is typically intractable for $t \geq 1$, which in turn makes it intractable to calculate the importance sampling weights $w_{\text{mar}}(x_T)$ for the T -th distribution.

In AIS, instead of attempting to calculate the importance sampling distribution corresponding to our original target distribution $\hat{\pi}$, we instead consider defining an

Algorithm 1 Annealed Importance Sampling (AIS)

Require: Sequence of unnormalized distributions $\{\hat{\pi}_t\}_{t=0}^T$ with $\hat{\pi}_T = \hat{\pi}$

Require: Forward transition kernels $\{F_t\}_{t=1}^T$ with F_t invariant to π_t

- 1: Sample $x_0^{(i)} \sim \pi_0(x)$ from initial distribution
 - 2: Initialize weight $w^{(i)} = 1$
 - 3: **for** $t = 1$ to T **do**
 - 4: Sample $x_t^{(i)} \sim F_t(x_t | x_{t-1}^{(i)})$ from forward transition
 - 5: Update weight $w^{(i)} \leftarrow w^{(i)} \cdot \frac{\hat{\pi}_t(x_{t-1}^{(i)})}{\hat{\pi}_{t-1}(x_{t-1}^{(i)})}$
 - 6: **end for**
 - 7: Set $x^{(i)} = x_T^{(i)}$ as final sample
 - 8: **return** $\{(x^{(i)}, w^{(i)})\}_{i=1}^N$
-

auxilliary target distribution over the extended space as follows

$$\Gamma(x_{0:T}) = \hat{\pi}(x_T) \prod_{t=0}^{T-1} B_t(x_t | x_{t+1}) \quad (2.21)$$

where $(B_t)_{t=0}^{T-1}$ are corresponding backwards Markov transition kernels that are invariant to π_t and transform intermediate samples x_t to x_{t+1} . For this extended space, it is straightforward to calculate the importance sampling weights similar to Equation (2.14) as

$$w(x_{0:T}) = \frac{\Gamma(x_{0:T})}{Q(x_{0:T})}. \quad (2.22)$$

These weights can be computed efficiently in an iterative manner while proceeding through the sampling procedure, as we see in Algorithm 1. Using these samples $x_0^{(i)}, \dots, x_T^{(i)}$ with their corresponding weights $w^{(i)}$, we can estimate expectations with Equation (2.16) and the partition function with Equation (2.15).

The AIS algorithm is quite flexible, and allows us to make a few choices in the design of the intermediate distributions $(\hat{\pi}_t)_{t=0}^T$ and the forward and backward transition kernels $(F_t)_{t=1}^T$ and $(B_t)_{t=0}^{T-1}$. For example, we can pick suitable forward transition kernels $(F_t)_{t=1}^T$ as long as they are invariant to the annealed distribution π_t , and common choices include MCMC kernels (Metropolis-Hastings, Gibbs', etc), Langevin dynamics, etc. Correspondingly, we can pick backwards transition kernels $(B_t)_{t=0}^{T-1}$ with properties beneficial to us. In fact, modern extensions of AIS allow for parametrised choices of F_t and B_t that can reduce the variance of the importance sampling weights, or be learnable through backpropagation, bringing them closer to our framework discussed in Equation (2.6).

In Neal (2001), Radford Neal chose a Metropolis-Hastings sampler along with a specific choice of backwards transition kernel given by

$$B_t^{\text{ais}}(x_t | x_{t+1}) = \frac{\hat{\pi}_{t+1}(x_t) F_{t+1}(x_{t+1} | x_t)}{\hat{\pi}_{t+1}(x_{t+1})}, \quad (2.23)$$

which simplifies the calculation of the importance sampling weights in Algorithm 1. Another closely-related algorithm is the Unadjusted Langevin Annealing (ULA) sampler (Wu et al., 2020; Welling et al., 2011; Thin et al., 2021), which defines the forward transition kernel as an approximate Langevin dynamics algorithm that uses intermediate annealed distributions at each time step (instead of the target distribution itself, as defined in Equation (2.17))

$$x_{t+1} \leftarrow x_t + \Delta t \nabla_x \log \hat{\pi}_t(x) + \sqrt{2\Delta t} \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I), \quad t = 0, \dots, T. \quad (2.24)$$

Furthermore, the backwards kernel is defined quite simply as the reverse of the forward kernel, i.e. $B_t^{\text{ula}}(x_t | x_{t+1}) = F_{t+1}(x_{t+1} | x_t)$.

Remark: Continuous Limit of AIS

We can see that the framework of AIS simplifies to the choice of a forward Markov Chain from our proposal μ to the target $\hat{\pi}$ parametrised by a forward transition kernel F_{θ_t} , and a backwards Markov Chain similarly parametrised by a backward transition kernel B_{ϕ_t} . These Markov chains converge to forwards and backwards SDEs in the continuous-time limit, similar to Equation (2.7). Furthermore, the effective proposal and target distributions converge to the corresponding path measures of the SDEs

$$Q(x_{0:T}) \rightarrow \overrightarrow{\mathbb{P}}_{\theta}, \quad \Gamma(x_{0:T}) \rightarrow \overleftarrow{\mathbb{P}}_{\phi} \quad (2.25)$$

Similarly, the importance sampling weights in Equation (2.22) converge to the density ratio between the path measures in Equation (2.25), which in turn is defined as a quantity called the Radon-Nikodym derivative

$$w(x_{0:T}) \rightarrow \frac{\overrightarrow{\mathbb{P}}_{\theta}(x_{0:T})}{\overleftarrow{\mathbb{P}}_{\phi}(x_{0:T})} \rightarrow \frac{d\overrightarrow{\mathbb{P}}_{\theta}}{d\overleftarrow{\mathbb{P}}_{\phi}} \quad (2.26)$$

We will see that the Radon-Nikodym derivative is not well-defined for the density ratio of a forward path measure w.r.t a backward path measure, and we will develop the theoretical framework to define this novel quantity in Chapter 4.

2.3 Generative Modelling

In certain situations, we don't have access to the explicit form of $\hat{\pi}$, or the ability to take gradients of the density. Instead, we have access to a collection of samples from the density through a training dataset $\mathcal{D}_{\text{train}} = \{x^{(i)}\}_{i=1}^{N_{\text{train}}}$. We wish to learn a parametrised approximation π_θ ² to the density of interest $\hat{\pi}$ for two goals: to draw new, novel samples from $\hat{\pi}$ and optionally, to be able to evaluate $\hat{\pi}$ on existing samples. This task is commonly called generative modeling, as we attempt to “generate” new samples from our unknown density by first “modeling” it through an existing training dataset. This paradigm has found immense success in the modern deep learning era, ranging from image synthesis to language modelling, as well as in scientific domains such as protein structure prediction and drug design.

In terms of our framework, many algorithms find it an easier task to first learn a mapping $p_\theta(x | z)$ from training samples $x \sim \mathcal{D}_{\text{train}}$ to a latent space z that has an easy-to-sample distribution μ . Given this mapping, many algorithms focus on different approaches to learn the “reverse” mapping, i.e. $q_\phi(z | x)$, allowing us to generate novel samples by sampling from μ and mapping it to the data distribution $\hat{\pi}$ through q_ϕ . The specific design choices of p_θ and q_ϕ cover a large family of algorithms, which we briefly discuss below. For the sake of completeness, we first discuss Variational Auto-Encoders (VAEs) (Kingma et al., 2019) as a flexible algorithm that follows our framework quite closely, albeit without a hierarchical structure by default. We then turn our attention to a large collection of algorithms that focus on designing more flexible mappings using hierarchical latent variables (whether implicitly or explicitly), such as Normalising Flows (NFs) (Papamakarios et al., 2021), Score Matching (SM) (Song and Ermon, 2019; Song, Sohl-Dickstein, et al., 2020) and Denoising Diffusion Probabilistic Models (DDPMs) (Ho et al., 2020).

²For notational clarity, we refer to the parametrised marginal probability distribution over x with π_θ , and the corresponding parametrised conditional probability distribution mapping z to x as p_θ .

2.3.1 Variational Auto-Encoders

VAEs parametrise p_θ (termed the *decoder*) from the latent space z to the data x with a neural network, allowing us to map individual datapoints from $x^{(i)} \sim \mathcal{D}_{\text{train}}$ to corresponding latent variables $z^{(i)}$. The issue arises when we attempt to recover an x corresponding to a sample $z \sim \mu$, as $p(z | x)$ is intractable due to the integral involved in the marginal likelihood term $p_\theta(x)$, as can be seen from Bayes' rule

$$p(z | x) = \frac{p_\theta(x | z)\mu(z)}{p_\theta(x)}, \quad p_\theta(x) = \int p_\theta(x | z)\mu(z)dz \quad (2.27)$$

Instead, a VAE learns a corresponding *encoder* model $q_\phi(z | x)$ to approximate $p(z | x)$. Here, we follow the somewhat unconventional derivation of Kingma et al. (2019) to derive the loss function for a VAE as a lower bound to the marginal log-likelihood, which is given as

$$\log p_\phi(x) = \mathbb{E}_{q_\theta(z|x)} [\log p_\phi(x)] \quad (2.28)$$

$$= \underbrace{\mathbb{E}_{q_\theta(z|x)} \left[\log \left[\frac{p_\phi(x, z)}{q_\theta(z | x)} \right] \right]}_{=\mathcal{L}_{\theta, \phi}(x) \text{ (ELBO)}} - \underbrace{\mathbb{E}_{q_\theta(z|x)} \left[\log \left[\frac{q_\theta(z | x)}{p_\phi(z | x)} \right] \right]}_{D_{KL}(q_\theta(z|x) || p_\phi(z|x))} \quad (2.29)$$

Since the KL-divergence term in Equation (2.29) is non-negative, we can see that the ELBO is a lower bound to the marginal log-likelihood, and in an ideal setting, one learns the parameters θ and ϕ by maximising the ELBO over the training dataset $\mathcal{D}_{\text{train}}$. In practice, VAEs often struggle to learn a sufficiently flexible encoder $q_\theta(z | x)$ and decoder $p_\phi(x | z)$, resulting in poor (or blurry) samples from the model. Intuitively, we can see how VAEs lie within our framework, albeit without using intermediate latent variables to parametrise the mapping. However, by allowing a more heirarchical structure, we can design more flexible models that can learn more complex mappings, which we will see in the next section.

2.3.2 Normalising Flows

Normalising flows learn the mapping from x to z using a series of composable invertible parametrised transformations $x = f_\theta(z)$, which provides the reverse mapping quite efficiently as the inverse of the function, giving us $z = f_\theta^{-1}(x)$. Then, using the change

of variables formula, we can write the probability distribution of x as

$$p_\theta(x \mid z) = \mu(z) \left| \det \nabla_z f_\theta^{-1}(z) \right|. \quad (2.30)$$

Normalising flows tend to perform really well in practise, owing to the flexibility of composing a series of iterative transformations to obtain a more complex mapping. However, the flexibility of these transformations is still limited by the requirements of invertibility and differentiability, which are often difficult to satisfy. As we see later, the idea of iterative mappings are central to the design of models we consider, and future methods have focused on designing more flexible transformations.

2.3.3 Score Matching

Energy-based approaches (such as EBMs) learn a parametrised representation of $\hat{\pi}$, and run into the issue of having to estimate the intractable normalising constant. On the other hand, Score Matching (SM) (Song and Ermon, 2019; Song, Sohl-Dickstein, et al., 2020) attempts to learn a different quantity, the “score” of the probability distribution. This quantity is defined as the gradient of the logarithm of the probability distribution, which gets rid of the necessity of learning the normalising constant,

$$s_\theta(x) \simeq \nabla_x \log \hat{\pi}(x) = -\nabla_x f_\theta(x) - \underbrace{\nabla_x Z_\theta}_0. \quad (2.31)$$

Vanilla score-matching models are usually learnt by minimising the Fisher divergence between the parametrised score function and the actual quantity of interest, giving us

$$\mathcal{L}_{\text{VSM}} = \mathbb{E}_{\hat{\pi}} \left[\|\nabla_x \log \hat{\pi}(x) - s_\theta(x)\|_2^2 \right]. \quad (2.32)$$

Due to the structure of the loss function taking an expectation over samples from $\hat{\pi}$, vanilla score-matching techniques often really struggle in approximating $\hat{\pi}$ in regions of low probability where samples are sparse. In order to address this issue, the authors resort to a technique that we will see repeated quite often in the current landscape: *iterative noising*. Instead of learning a score model for $\hat{\pi}$ directly, score-matching creates noisier and noisier versions of $\hat{\pi}$ by adding successively larger amounts of Gaussian noise, and learns a score model per level of noise. If we assume a schedule of noise values $\sigma_1 \leq \dots \leq \sigma_T$, we can write down each noised distribution as

$$\pi_t(x) = \int \hat{\pi}(y) \mathcal{N}(x; y, \sigma_t^2) dy, \quad t = 1, \dots, T. \quad (2.33)$$

We can then modify our score model to learn a different parametrisation for a different noise scale, resulting in a Noise-Conditional Score Network (NCSN) given by $s_\theta(x, t)$ and a weighted sum of Fisher divergences per noise scale, given as

$$\mathcal{L}_{\text{SM}} = \sum_{t=1}^T \mathbb{E}_{\pi_{\sigma_t}} \left[\|\nabla_x \log \pi_{\sigma_t}(x) - s_\theta(x, t)\|_2^2 \right]. \quad (2.34)$$

While score-matching was initially motivated by this heuristic argument about learning the score of a multi-level noised distribution, in Song, Sohl-Dickstein, et al. (2020), the authors show that the noising process can be represented as a stochastic differential equation that iteratively noises samples from the data distribution $\hat{\pi}$ until they resemble Gaussian noise (i.e. μ).

Remark: Continuous Limit of Score-Matching

As shown in (Song, Sohl-Dickstein, et al., 2020), the iterative noising process can be represented as a stochastic differential equation in the continuous limit. Here, we redefine our notation to follow standard notation in the literature closely, and assume that the distribution at $t = 0$ corresponds to the data distribution $\hat{\pi}$, and the distribution at $t = T$ corresponds to the base distribution μ . In this case, the iterative noising process can be represented as a forward SDE, transporting samples from $\hat{\pi}$ to μ , and is given by

$$dX_t = \sqrt{\frac{d\sigma_t^2}{dt}} dW_t, \quad X_0 \sim \hat{\pi}, \quad t = 0, \dots, T, \quad (2.35)$$

where W_t is a Wiener process^a, and with corresponding path measure $\vec{\mathbb{P}}_\phi$, where $\phi = \{\sigma_t\}$. As we will see in Chapter 3, it is possible to define a “reversal” of an SDE that shares the same marginal probability distributions as the original SDE, but travels in the opposite direction. In this case, the “reversal” of the backward SDE in Equation (2.35) will be a backward SDE that transports the base distribution μ to the data distribution $\hat{\pi}$, is parametrised by the variable θ through the inclusion of a score term in backward drift, and has a corresponding path measure $\vec{\mathbb{P}}_\theta$. As we will show more concretely in Chapter 4, the loss function for score-matching as defined in Equation (2.34) converges to minimising the KL divergence between the path measures $\vec{\mathbb{P}}_\phi$ and $\vec{\mathbb{P}}_\theta$.

^aWe define these processes more rigorously in Chapter 3.

2.3.4 Denoising Diffusion Probabilistic Models

Denoising Diffusion Probabilistic Models (DDPMs) (Ho et al., 2020) are another popular framework for generative modelling using the notion of “iterative noising” that have found massive success. DDPM imposes additional structure on the mapping from data to latent $q_\phi(z | x)$ by introducing a series of additional latent variables $\{x_1, \dots, x_{T-1}\}$ with $x_0 := x$ and $x_T := z^3$, and assuming a Markov assumption on the chain of latents, such that x_t given x_{t-1} is independent of previous latents. The authors then specifically choose a form of “noising” process that iteratively adds Gaussian noise to each latent at time $t - 1$ to obtain the latent at time t , giving us the noising distributions

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I), \quad (2.36)$$

for some choice of noising β_t schedule. Due to each individual noising process being Gaussian, a nice property of the DDPM formulation is that we can represent the latent at any time step t w.r.t the original data sample $x = x_0$ in closed-form⁴, giving us

$$q(x_t | x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I\right), \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s \text{ with } \alpha_t = 1 - \beta_t. \quad (2.37)$$

If we could now learn a reverse “denoising” process $p(x_{t-1} | x_t)$ that takes a sample from the noising distribution at time t and maps it back to the original data distribution iteratively, we could sample from the original data distribution by starting with a sample from the base distribution $x_T \sim \pi_0$ and iteratively applying the reverse noising process. However, calculating the exact reverse noising process is intractable, as we see here

$$p(x_{t-1} | x_t) = \frac{q(x_t | x_{t-1})p(x_{t-1})}{p(x_t)} \quad (2.38)$$

$$= \frac{q(x_t | x_{t-1}) \int q(x_{t-1} | x_0)\pi_0(x_0)dx_0}{\int q(x_t | x_0)\pi_0(x_0)dx_0}, \quad (2.39)$$

where the integrals over the data distribution π_0 are intractable. Instead, DDPM assumes a variational distribution $p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$ that is a Gaussian with mean and variance parametrised by a neural network, and learns the parameters θ in the same way as a VAE (where the latents $z = \{x_1, \dots, x_T\}$). We can

³We note here that this convention is the opposite of that considered by sampling techniques such as AIS, but consistent with the notation chosen by Score Matching algorithms, as can be seen from Equation (2.35).

⁴See Appendix A for a proof.

therefore write down an ELBO similar to Equation (2.29), giving us

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{x_{1:T} \sim q(x_{1:T} | x_0)} \left[\log \left[\frac{p_\theta(x_{1:T} | x_0)}{q(x_{1:T} | x_0)} \right] \right]. \quad (2.40)$$

Interestingly, Song, Sohl-Dickstein, et al. (2020) showed that in the continuous limit, the iterative noising process for DDPM converges to an SDE (different from the SDE in Score Matching), and therefore both Score Matching and DDPM are variants of a score-matching loss, albeit on different forward SDEs.

Remark: Continuous Limit of DDPM

As shown in Song, Sohl-Dickstein, et al. (2020), the iterative noising process in Equation (2.36) can be represented as a stochastic differential equation in the continuous limit. Making the same assumptions as Score Matching ($t = 0$ corresponds to the data distribution $\hat{\pi}$ and $t = T$ corresponds to the base distribution μ), the forward SDE is given by

$$dX_t = -\frac{1}{2}\beta_t X_t dt + \sqrt{\beta_t} dW_t. \quad (2.41)$$

The DDPM objective, which is equivalent to the Score Matching objective (Song, Sohl-Dickstein, et al., 2020), can therefore be written in terms of the KL divergence between the forward and backward path measures.

2.4 Conditional Sampling

In the conditional sampling framework, we often assume one of two situations. First, in a generative modelling setting, we have access to the unconditional distribution $\hat{\pi}$ through the mapping $p_\theta(x | z)$, and we wish to instead learn a mapping to a modified distribution $\pi_{\cdot|y}$ that is a “tilted” or “conditioned” version of $\hat{\pi}$ based on some additional constraints. We can write this additional constraint as either a generic reward function $r(x)$ that results in $\pi_{\cdot|y} \propto \hat{\pi} \exp(r(x))$, or an explicit likelihood model $p(y | x)$ that results in $\pi_{\cdot|y} \propto \hat{\pi} p(y | x)$. In this situation, we attempt to learn a conditional mapping $p_{\cdot|y}(x | y, z)$ that modifies the original mapping $p_\theta(x | z)$ based on a dataset of datapoints and conditions $\mathcal{D}_{\text{cond}} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N_{\text{cond}}}$. This approach is termed *conditional generative modelling*, as we eventually learn a way to map from a latent space to the conditioned distribution, often with the (somewhat strong) assumption that we have a-priori access to a map to the unconditioned distribution.

Similar to our previous discussions, many cases of generative modeling algorithms that converge to stochastic differential equations allow for additional conditioning that also has a continuous time perspective in the SDE regime. In fact, a large number of baseline algorithms that modify existing iterative noising generative modeling algorithms such as Score Matching (Section 2.3.3) and DDPM (Section 2.3.4) are easier to understand with the SDE perspective, and so we defer their discussion until the next Chapter.

The second situation arises in more specific situations where we have the ability to efficiently draw samples from the unconditional distribution $\hat{\pi}$, and we wish to transform these samples by modifying them to instead have been generated from the conditioned distribution $\pi_{\cdot|y}$. This approach is called *pathwise conditioning*. There are often drawbacks in this situation: we may not be able to learn an arbitrary map that can map latent draws to samples from the conditional distribution, and only be able to convert samples from the unconditional distribution to ones from the conditional distribution. Also, this is usually computationally efficient only when drawing samples from the unconditional distribution are feasible or efficient in the first place, a situation that is quite rare in high-dimensional distributions. A very useful family of distributions that are quite appealing given this framework are Gaussian distributions, where it is often easy to sample from the unconditional distribution, and under specific likelihood models, the conditional distribution is also Gaussian, allowing for closed-form calculation of the correction factor using Matheron’s rule. In the limit of Gaussian distributions on an infinite number of datapoints, we converge to a special case of stochastic process termed a Gaussian Process (GP). In the second half of this thesis, we will extensively develop the theory of GPs, and show how they shine in particular applications where uncertainty estimation is valuable: Bayesian inference. We will show how many Bayesian inference problems in the deep learning setting can be simplified to GPs, and therefore GPs provide a powerful, unifying framework for a scalable, flexible, and high-performance Bayesian inference framework.

Chapter 3

Stochastic Processes for Sampling

In the previous chapter, we discussed a large collection of sampling algorithms, and alluded to the fact that they converge to stochastic differential equations in the continuous-time limit. As we discussed in Equation (2.18), we are interested in differential equations of the form

$$\frac{dX_t}{dt} = \mu(X_t) + \sigma(X_t) \times \text{“noise”}, \quad (3.1)$$

where the equation is akin to an ordinary differential equation (ODE) with a drift parameter μ , and random noise injected into it at each time step scaled by a diffusion parameter σ . As a result, the solution to this equation is no longer deterministic, but a random variable at each time step. This means that while the explicit value of X_t is not known at each time step without knowing the initial condition X_0 as well as the noise, we can always write down the probability distribution $\mathbb{P}(X_t)$ at each time step. In this chapter, we will develop the mathematical machinery required to discuss stochastic processes and SDEs. We will then briefly show how some of our implied connections from the previous chapter arise naturally from simple design choices in the SDE framework. Finally, we highlight drawbacks of the existing SDE framework, preventing the development of a unified framework, allowing us to tackle this task in the upcoming chapters.

3.1 A Review of Probability Theory

The stochastic calculus framework is built on measure-theoretic probability, which we describe in this section. We build upon the notation and definitions introduced in

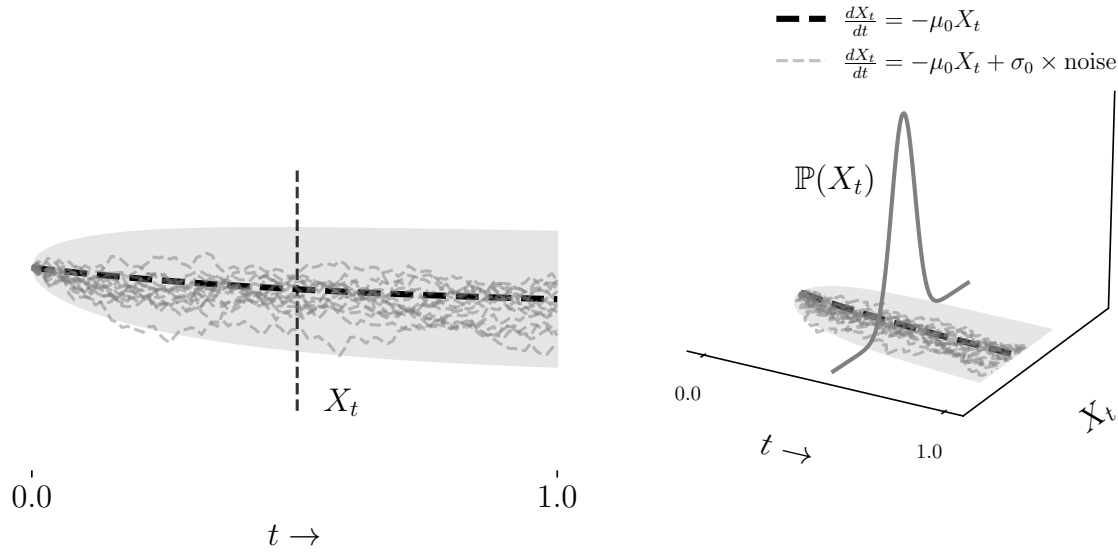


Fig. 3.1. A simulation of an ODE (--) and multiple realisations of an SDE (---) with drift parameter $-\mu_0 X_t$ and diffusion parameter $\sigma_0 \times \text{noise}$. At each time step, the ODE takes a deterministic step, while the SDE takes a random step. At $t = 0.5$, the random variable X_t will have unique values determined by the intersection of the dotted black line with the realisations of the SDE. On the right, we plot the probability distribution $\mathbb{P}(X_t)$ at $t = 0.5$, which is a Gaussian distribution for this specific choice of parameters.

Oksendal (2013), which we refer the reader to for a more thorough treatment. Instead, we aim to provide a concise summary of the most relevant concepts, along with an intuitive understanding of the concepts.

At its core, measure-theoretic probability theory is concerned with the notion of assigning a “measure” to subsets of a space. This notion gets more complicated in higher dimensions (or infinite dimensions), or when we consider arbitrary collections of sets, and so we aim to be as concrete as possible. Therefore, let us put a few definitions to these abstract notions.

Probability Measure and Probability Space. We consider the *sample space* Ω , which is the set of all possible outcomes ω of an experiment. When an experiment is performed, the outcome is some element of Ω based on a law of probability $\mathbb{P}$. We then consider an *event* A to be a subset of Ω , which is a collection of some of the possible outcomes. It is useful to restrict any theoretical considerations we have to the σ -algebra $\mathcal{F}$, which is a collection that contains Ω along with all subsets of Ω that are closed under countable unions, intersections, and complements. In general, the σ -algebra is a subset of the power set $\mathcal{P}(\Omega)$, as there are often cases when the power set can contain

pathological sets that are not measurable. We define a *probability measure* as the map $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ that satisfies the following properties:

$$(P0) \quad \mathbb{P}(\emptyset) = 0 \quad (3.2)$$

$$(P1) \quad \mathbb{P}(\Omega) = 1 \quad (3.3)$$

$$(P2) \quad \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i) \quad \text{for all } A_i \in \mathcal{F}. \quad (3.4)$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a *probability space*. Further, we say that an event $A \in \mathcal{F}$ happens *almost surely* (*a.s.*) if $\mathbb{P}(A) = 1$ ¹.

Borel σ -algebra and Lebesgue Measure. In the special case when we consider Ω to be the real space $\mathbb{R}^n$ (which is one example of a topological space), we further restrict our attention to the Borel σ -algebra $\mathcal{B}(\mathbb{R}^n)$, which is the smallest σ -algebra that contains all open sets of $\mathbb{R}^n$. Correspondingly, we can define the “canonical” measure on $\mathcal{B}(\mathbb{R}^n)$, known as the Lebesgue measure λ , which assigns a measure to each Borel set.

Random Variables. Probability spaces give us a way to assign probabilities to collections of interesting subsets of Ω . However, we don’t often work with the sets themselves, which can be hard to visualise or enumerate, but functions that map these sets (or events) to the real line. We define a *$\mathcal{F}$ -measurable function* $X : \Omega \rightarrow \mathbb{R}^n$ to be a function that satisfies $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R}^n)$. We also call such functions *random variables*².

Change of Measure and the Radon-Nikodym Theorem. On a given measurable space $(\Omega, \mathcal{F})$, if we consider two measures μ and ν such that they are *absolutely continuous* with respect to each other (i.e. $\nu \ll \mu$), we can express ν in terms of μ using a change of measure. Intuitively, we could think of this as a process of assigning a “weight” to each element of Ω such that when we take the weighted sum of elements over a measurable subset $A \in \mathcal{F}$ with measure $\mu(A)$, we get the measure of A under ν . That is, there exists a function $f : \Omega \rightarrow \mathbb{R}$ such that $\nu(A) = \int_A f d\mu$. This function f is known as the *Radon-Nikodym derivative* of ν with respect to μ , and is denoted by $\frac{d\nu}{d\mu}$.

Pushforward Measure. Considering two measurable spaces $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$, if we have a measurable function $f : \Omega_1 \rightarrow \Omega_2$ and a measure μ on Ω_1 , we can define

¹Note that this is not the same as saying that $A = \Omega$, as there may be elements in Ω that are not in A , but this set of elements has measure zero.

²A somewhat unfortunate choice of terminology, as random variables are not variables at all, but rather functions that map the sample space to the real line.

a new measure ν on Ω_2 by $\nu(A) = \mu(f^{-1}(A)), \forall A \in \mathcal{F}_2$. This measure ν is known as the *pushforward measure* of μ by f , and is denoted by $f_*\mu$. We can think of this as a process of “pushing” the measure μ through the function f to define a new measure on Ω_2 .

Probability Distributions and Probability Density Functions. Since random variables are functions that map the space $(\Omega, \mathcal{F}, \mathbb{P})$ to the real line $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, there exists a pushforward measure of $\mathbb{P}$ through X , which we write as $X_*\mathbb{P}$. This pushforward measure is termed the *probability distribution* of X , and is denoted by $\mathbb{P}_X$. Since $\mathbb{P}_X$ is a measure defined on $\mathbb{R}^n$, we can further define the Radon-Nikodym derivative of $\mathbb{P}_X$ with respect to the Lebesgue measure λ on $\mathbb{R}^n$, which is known as the *probability density function* of X , and is denoted by f_X .

Definition 3.1 (Stochastic Process). A stochastic process $\{X_t\}_{t \in T}$ is a parametric collection of random variables indexed by a parameter $t \in T$, which we call the *index set*.

While we can define stochastic processes on arbitrary index sets, we are particularly interested in the case when T is the time index $t \in [0, \infty)$. Later on, when we discuss Gaussian processes, we will see that the index set can also be a continuous domain, such as the real line.

Definition 3.2 (Filtration and Filtered Probability Space). A filtration of the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a sequence of sigma algebras $\{\mathcal{F}_t\}_{t \in T}$ satisfying $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ if $s \leq t$. We call a filtered probability space the tuple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in T}, \mathbb{P})$.

Intuitively, a filtration is a way of representing the information available at each time t . We can study the properties of the stochastic process X_t at a specific time t by working within the σ -algebra $\mathcal{F}_t$, which represents the information available up to that time. As time passes, more information becomes available, and so the σ -algebra $\mathcal{F}_t$ becomes larger as t increases.

Definition 3.3 (Adapted and Predictable Processes). Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in T}, \mathbb{P})$ be a filtered probability space. A stochastic process $\{X_t\}_{t \in T}$ is $\mathcal{F}_t$ -adapted if X_t is $\mathcal{F}_t$ -measurable, and is $\mathcal{F}_t$ -predictable if X_t is $\mathcal{F}_{t-}$ -measurable.

We can also generate a new filtered probability space given a stochastic process $\{X_t\}_{t \in T}$, by considering the filtration generated by the process, which is the smallest filtration that makes the process adapted.

Definition 3.4 (Filtration Generated by a Process). Let $\{X_t\}_{t \in \mathcal{T}}$ be a stochastic process. The filtration generated by $\{X_t\}_{t \in \mathcal{T}}$ is defined as $\mathcal{F}_t^X = \sigma\{X_0, \dots, X_t\}$, and the process $\{X_t\}_{t \in \mathcal{T}}$ is $\mathcal{F}_n^X$ -adapted by construction.

Further, we will consider a special family of stochastic processes that have very useful properties, and pop up often in our further discussions: martingales.

Definition 3.5 (Martingale). A stochastic process X_t is a martingale with respect to a filtration $\mathcal{F}_t$ if the following properties hold:

$$(P1) \quad X_t \text{ is } \mathcal{F}_t \text{ measurable for all } t \in \mathcal{T} \text{ (it is adapted)} \quad (3.5)$$

$$(P2) \quad \mathbb{E}[|X_t|] < \infty \text{ for all } t \in \mathcal{T} \text{ (it is integrable)} \quad (3.6)$$

$$(P3) \quad \mathbb{E}[X_t | \mathcal{F}_s] = X_s \text{ a.s. for all } s \leq t. \quad (3.7)$$

Remark: An Intuitive View of Martingales

The word (and really the concept) of a martingale originally came from game theory and gambling, where a martingale referred to a betting strategy where the present is the best approximation of the future, and on average, you would not win or lose money, but stay the same. In the context of stochastic processes, a martingale is a stochastic process that is a “fair” process, in the sense that it neither grows nor decays on average. This is quite easily seen through Equation (3.7), where the expected value of the process at time t , given only information up to time s , is exactly the value of the process at time s .

Another useful property of a martingale for us to remember for later is that it is a stochastic process where the optimal regressor of a future value conditioned on the past is the past value itself. We first define the optimal predictor of mean-squared error

Theorem 3.6 (Optimal Predictor of Mean-Squared Error). Let X, Y be $\mathcal{L}^2$ random variables and

$$f^* = \arg \min_{f \text{ measurable}} \mathbb{E}_{X,Y} [\|f(X) - Y\|^2]. \quad (3.8)$$

Then it follows that $f^*(x) = \mathbb{E}[Y | X = x]$.

Applying the optimal predictor of a mean-squared error theorem to the martingale, gives us the following corollary

Corollary 3.7 (Martingale as Optimal Predictor).

$$f^* = \arg \min_{f-\text{measurable}} \mathbb{E}[\|f(X_t) - X_{t+\Delta t}\|^2], \quad f^*(X_t) = \mathbb{E}[X_{t+\Delta t} \mid X_t] = X_t. \quad (3.9)$$

3.2 A Discrete-Time Intuition for SDEs

Now that we have introduced the necessary mathematical framework, let us revisit the SDE in Equation (3.1). The main challenge in this differential equation is the inability to concretely parameterise the noise term. To address this, let us consider a discrete-time approximation of this equation with a simple choice of $\mu = 0$ and $\sigma = 1$. If we discretise the time interval $[0, 1]$ into N intervals of size $\Delta t = \frac{1}{N}$, we can approximate the solution to the SDE at time $t = 1$ as

$$X_1 = X_0 + \sum_{i=1}^N Z_i, \quad (3.10)$$

where Z_i refers to the noise term at each time step. We need to decide a few properties of the noise term Z_i . First, we assume that the noise terms are independent and identically distributed (i.i.d.). Second, we need to decide the distribution of the noise terms. However, anytime we are faced with a sum over i.i.d. random variables, we can rely on the Central Limit Theorem to tell us that the distribution of the sum will be Gaussian. If we make some reasonable assumptions about the distribution of Z_i , i.e. specify the first and second moments as $\mathbb{E}[Z_i] = \nu N^{-1}$ and $\mathbb{V}[Z_i] = \zeta^2 N^{-1}$, we have

$$X_1 - X_0 \approx \mathcal{N}(\nu, \zeta^2). \quad (3.11)$$

We extend this result to a general time t by dividing each unit interval into N subintervals, giving us the difference between any X_t and X_s (for $s \leq t$) as

$$X_t - X_s \sim \mathcal{N}(\nu(t-s), \zeta^2(t-s)). \quad (3.12)$$

Definition 3.8 (Discrete-Time Noise Process). *We define the discrete-time noise process $\{X_t\}_{t \in T}$ as the collection of random variables that satisfies the following properties:*

1. *Normally-distributed increments:* $X_t - X_s \sim \mathcal{N}(\nu(t-s), \zeta^2(t-s)).$

2. *Independent increments:* $X_t - X_s$ is independent of $X_u - X_v$ for any s, t, u, v such that $0 \leq s < t \leq u < v$.

Our discrete-time approximation of the SDE shows us that the noise term has these two properties. In order to extend this result to the continuous-time limit, where we can say that any collection of random variables $\{X_t\}_{t \in T}$ has these properties over the continuous domain of time, we use the following theorem:

Theorem 3.9 (Kolmogorov's Continuity Criterion). *If $\{X_t\}_{t \in T}$ is a real-valued process such that for all $T > 0$ there exist constants $\alpha, \beta, \gamma > 0$ with*

$$\mathbb{E}[|X_{t+h} - X_t|^\alpha] \leq \gamma h^{1+\beta}. \quad (3.13)$$

for every t and $0 \leq h \leq T$, then there exists a modification of X_t which has almost-surely continuous paths.

Proof. See Stroock et al. (1997, Pg. 51). ■

Using Theorem 3.9, we can now extend our discrete-time approximation of the noise term in Definition 3.8 to the continuous-time limit.

Proposition 3.10 (Continuous Modification of Noise). *There exists a modification of the stochastic process $\{X_t\}_{t \in T}$ defined in Definition 3.8 that has almost-surely continuous paths.*

Proof. We can use Theorem 3.9 to show that there exists a modification of the stochastic process $\{X_t\}_{t \in T}$ that has almost-surely continuous paths. From Equation (3.12), we know that $X_{t+h} - X_t \sim \mathcal{N}(\nu h, \xi^2 h)$. We calculate moments of the random variable $X_{t+h} - X_t$ using the moment generating function of the normal distribution, and find for the fourth moment that

$$\mathbb{E}[|X_{t+h} - X_t|^4] = M''''(0) \quad (3.14)$$

$$= \nu^4 h^4 + \nu^3 \xi^2 h^3 + 3\xi^4 h^2 \quad (3.15)$$

$$= (\nu^4 h^2 + \nu^3 \xi^2 h + 3\xi^4) h^2 \quad (3.16)$$

$$\leq (\nu^4 T^2 + \nu^3 \xi^2 T + 3\xi^4) h^2. \quad (3.17)$$

Therefore, for $\alpha = 4$, $\beta = 1$ and $\gamma = \nu^4 T^2 + \nu^3 \xi^2 T + 3\xi^4$, we satisfy Theorem 3.9, and so there exists a modification of the stochastic process $\{X_t\}_{t \in T}$ that has almost-surely continuous paths. ■

Using Proposition 3.10, we can now extend our discrete-time approximation of the noise term in Definition 3.8 to the continuous-time limit. This modification is known as the Wiener process, or Brownian motion, and is denoted by W_t .

Definition 3.11 (Brownian Motion). *A real-valued stochastic process $\{W_t\}_{t \geq 0}$ is a (standard) Brownian motion starting at $x \in \mathbb{R}$ if:*

$$(P0) \quad W_0 = x \tag{3.18}$$

$$(P1) \quad W_t - W_s \sim N(0, t - s) \tag{3.19}$$

$$(P2) \quad \text{if } 0 = t_0 \leq t_1 \leq t_2 \leq \dots \leq t_n \text{ then} \tag{3.20}$$

$$W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}} \text{ are independent,} \tag{3.21}$$

$$(P3) \quad t \mapsto W_t \text{ is continuous, almost surely.} \tag{3.22}$$

Further, as we were doing with stochastic processes before, we can consider a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, \infty)}, \mathbb{P})$ where W_t is adapted to the filtration $\{\mathcal{F}_t\}_{t \in [0, \infty)}$. We have the following nice property of Brownian Motion:

Lemma 3.12. *Adapted Wiener Process is a Martingale The $\mathcal{F}_t$ -adapted Wiener process W_t is a martingale with respect to the filtration $\{\mathcal{F}_t\}_{t \in [0, \infty)}$.*

Proof. In order to prove this, we must show that $\mathbb{E}(W_t \mid \mathcal{F}_s) = W_s$ for any $t > s$. But as W_s is $\mathcal{F}_s$ -measurable (by adaptedness) this is equivalent to $\mathbb{E}(W_t - W_s \mid \mathcal{F}_s) = 0$, and this is clearly true by the definition of the Wiener process (as $W_t - W_s$ has zero mean and is independent of $\mathcal{F}_s$). ■

Intuitively, we can see that Brownian Motion is the continuous limit of variables that are represented as sums of i.i.d. normal random variables. Given our original assumptions about the i.i.d.ness of Z_k , we can see that in the continuous limit of $\delta t \rightarrow 0$, we have

$$\sum_{s_i \leq t} Z_{s_i} \rightarrow \int_{s \leq t} Z_s \, ds \approx W_t. \tag{3.23}$$

Therefore, the noise would ideally be defined as some notion of the “derivative” of Brownian Motion dW_t , giving us the SDE from Equation (3.1) as

$$\frac{dX_t}{dt} = \mu(X_t) + \sigma(X_t) \times dW_t. \tag{3.24}$$

Unfortunately, we hit a roadblock when we try to define the noise as dW_t . The Wiener process is not differentiable in the traditional sense, and so we need to define a new notion of the derivative of the Wiener process. This is where Itô calculus comes in.

sp: Maybe cover unbounded variation of B.M, and $(dW)^2 = dt$

3.3 Itô Calculus

Let us rewrite Equation (3.24) as an integral equation, giving us

$$X_t = X_0 + \int_0^t \mu(X_s) ds + \int_0^t \sigma(X_s) dW_s \quad (3.25)$$

The first term on the right-hand side is a Riemann-Stieltjes integral, and can be defined using standard rules of integration. The second term, however, is undefined in the traditional sense, since the integrand itself is a “noisy process” and has unbound total variation. As a simple example, consider discretising the integral of Brownian Motion itself over the interval $[0, t]$, giving us

$$\int_0^t W_s dW_s \approx \sum_{k=0}^{N-1} W_{t_k^*} [W_{t_{k+1}} - W_{t_k}], \quad t_k^* \in [t_k, t_{k+1}]. \quad (3.26)$$

For a Riemann-Stieltjes integral, we can pick any point t_k^* in the interval $[t_k, t_{k+1}]$ to approximate the integral, as long as the upper and lower Darboux sums converge to the same value. However, with some simple algebra, we can see that this is not the case. Considering $t_k^* = t_k$, we have

$$\int_0^t W_s dW_s \approx \sum_{k=0}^{N-1} W_{t_k} [W_{t_{k+1}} - W_{t_k}] \quad (3.27)$$

$$= \frac{1}{2} W_t^2 - \frac{1}{2} W_0^2 - \frac{1}{2} \langle W \rangle_t \quad (3.28)$$

$$= \frac{1}{2} W_t^2 - \frac{1}{2} W_0^2 - \frac{t}{2}. \quad (3.29)$$

Instead, if we “peak” into the future and consider $t_k^* = t_{k+1}$, we have

$$\int_0^t W_s dW_s \approx \sum_{k=0}^{N-1} W_{t_{k+1}} [W_{t_{k+1}} - W_{t_k}] \quad (3.30)$$

$$= \frac{1}{2} W_t^2 - \frac{1}{2} W_0^2 + \frac{t}{2}. \quad (3.31)$$

This mismatch raises a big issue with how we might be able to define stochastic integrals, as it is no longer the case that the partitioned sums should converge to a single value regardless of the choice of t_k^* . In fact, different choices to t_k^* result in the corresponding processes satisfying different fundamental properties as well. the first choice of $t_k^* = t_k$ results in the integral being an *Itô integral*, and the second choice of $t_k^* = t_{k+1}$ results in a *Stratanovich integral*.

3.3.1 Properties of Itô Integrals

The choice of $t_k^* = t_k$ equips the Itô integral with a number of desirable properties, which we discuss in this section. The most important

3.3.2 Construction of Itô Integrals

We have defined the Itô integral for Brownian motion, having decided which point in an interval to pick (i.e. the leftmost). In order to extend the definition of the Itô integral to more general stochastic processes, we can follow the following steps:

1. Defining the Itô integral for simple step-wise processes
2. Proving the Itô isometry for simple processes
3. Constructing a sequence of simple processes that converge to a general stochastic process in a mean-square sense
4. Defining the Itô integral as the limit of the sequence of simple processes

For the sake of brevity, we refer to Oksendal (2013, Pg. 21, Section 3.1) for this construction, choosing here to state the important properties and definitions, namely the Itô isometry and the Itô integral. First, we define the family of processes that we will consider, namely processes that are bounded and adapted.

Definition 3.13 (Class of Bounded and Adapted Processes). *Let $\mathcal{V} = \mathcal{V}(S, T)$ be the class of functions*

$$f(t, \omega) : [0, \infty) \times \Omega \rightarrow \mathbf{R} \quad (3.32)$$

such that (i) $(t, \omega) \rightarrow f(t, \omega)$ is $\mathcal{B} \times \mathcal{F}$ -measurable, where $\mathcal{B}$ denotes the Borel σ -algebra on $[0, \infty)$. (ii) $f(t, \omega)$ is $\mathcal{F}_t$ -adapted. (iii) $E \left[\int_S^T f(t, \omega)^2 dt \right] < \infty$.

Definition 3.14 (Itô Integral). Let V_t be a bounded and adapted process as per Definition 3.13. Then there exists a sequence of simple processes $(V_t^n)_{n=1}^\infty$ such that

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\int_0^T (V_u - V_u^n)^2 du \right] = 0 \quad (3.33)$$

The Itô integral then exists (in $\mathcal{L}^2(\mathbb{P})$) and is defined as

$$I := \lim_{n \rightarrow \infty} \int_0^T V_u^n dW_u, \quad (3.34)$$

which is independent of the choice of the sequence $(V_t^n)_{n=1}^\infty$.

Finally, we state the Itô isometry, which is a key property of the Itô integral.

Definition 3.15 (Itô Isometry).

$$E \left[\left(\int_S^T f(t, \omega) dB_t \right)^2 \right] = E \left[\int_S^T f^2(t, \omega) dt \right] \quad \text{for all } f \in \mathcal{V}(S, T). \quad (3.35)$$

3.3.3 Itô Processes and Itô's Lemma

Now that we've been able to define the Itô integral $\int V_t dW_t$ for more general stochastic processes, we can consider other properties of calculus that are essential for any meaningful analysis, such as the chain rule. Given a stochastic process defined as

$$X_t = X_0 + \int_0^t a_s(X_s) ds + \int_0^t V_s dW_s \quad (3.36)$$

or as a stochastic differential equation,

$$dX_t = a_t(X_t) dt + V_t dW_t \quad (3.37)$$

X_t here is itself a stochastic process defined as a combination of an Itô integral and an ordinary integral, and we term it an *Itô process*³.

Can we calculate what the stochastic process $f(X_t)$ would look like? Yes, in fact it can be shown that there exists a version of chain rule for stochastic processes, and it is

³The term “Itô process” is used to describe a stochastic process that can be expressed as an Itô integral. Frequently, the term is used interchangeably with “stochastic integral” or “stochastic process”.

known as Itô's lemma. First, we consider a simple version, where we have $X_t = W_t$ and $a_t = 0$. Then we have

Definition 3.16 (Stochastic Chain Rule). *Suppose $f \in C^2([0, \infty) \times \mathbb{R})$ (i.e. $f(t, x)$ is twice continuously differentiable in x). Then $Y_t := f(W_t)$ is an Itô process, and*

$$dY_t = \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(W_t) dt + \frac{\partial f}{\partial x}(W_t) dW_t \quad (3.38)$$

We can extend this to the general case where X_t is an Itô process given by Equation (3.37), and $f \in C^2([0, \infty) \times \mathbb{R})$. In fact, the proof will follow quite similarly as above, where we need to make sense of convergence in mean and in probability, and apply the same principles. The more popular (and more general) version of Itô's lemma is given below.

Theorem 3.17 (Itô's Lemma). *Let X_t be an Itô process given by Equation (3.37), and $f \in C^2([0, \infty) \times \mathbb{R})$. Then $Y_t := f(X_t)$ is an Itô process, and is defined as*

$$dY_t = \left(\frac{\partial f}{\partial t}(t, X_t) + a_t \frac{\partial f}{\partial x}(t, X_t) + \frac{1}{2} V_t^2 \frac{\partial^2 f}{\partial x^2}(t, X_t) \right) dt + V_t \frac{\partial f}{\partial x}(t, X_t) dW_t. \quad (3.39)$$

Alternatively, we can write this as

$$dY_t = \frac{\partial f}{\partial t}(t, X_t) dt + \frac{\partial f}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) (dX_t)^2 \quad (3.40)$$

3.4 Path Measures and their Laws of Probability

We can now write down the solution to the SDE in Equation (3.25) as

$$dX_t = \mu(X_t) dt + \sigma(X_t) dW_t, \quad (3.41)$$

where it can be readily seen that the solution is an Itô process. If we were to consider the probability distribution over a discrete subset of random variables indexed by $t_0, t_1, \dots, t_n$ along the “path” of this process, we could write the joint distribution as

$$\mathbb{P}(X_{t_0} \in A_0, X_{t_1} \in A_1, \dots, X_{t_n} \in A_n), \quad (3.42)$$

where $A_i \subset \mathbb{R}$ is a Borel set. This is known as the law of the process. In the continuous limit, where we have a (potentially) infinite number of random variables indexed by

functions belonging to $\mathcal{T} = C([0, T], \mathbb{R}^n)$, we define the corresponding probability measures as *path measures*.

Definition 3.18 (Path Measures). *Let $\mathcal{T} = C([0, T], \mathbb{R}^n)$ be the space of continuous functions from $[0, T]$ to $\mathbb{R}^n$, and $\mathcal{B}(\mathcal{T})$ be the Borel σ -algebra of $\mathcal{T}$. Then a path measure is a probability measure on the space $(\mathcal{T}, \mathcal{B}(\mathcal{T}))$.*

Therefore, for solutions of SDEs of the form in Equation (3.25), we can define the *law of the process* as a path measure, notated as $\mathbb{P} = \text{Law}(\{X_t\}_{t \in T})$.

Definition 3.19 (Path Measure of an Itô Process). *Let $\{X_t\}_{t \in T}$ be an Itô process defined as the solution to the SDE in Equation (3.25), with drift coefficient μ and diffusion coefficient σ . Further, let π_0 be the marginal distribution of the RV at $t = 0$. Then the path measure of $\{X_t\}_{t \in T}$ is denoted as $\mathbb{P}^{\mu, \pi_0} = \text{Law}(\{X_t\}_{t \in T})$.*

In this section, we discuss certain properties of path measures that could be considered “laws of probability” for operations on paths of an Itô process. These properties are often continuous-time analogues of the standard laws of probability, and will be very useful in the upcoming chapters.

3.4.1 Disintegration Theorem: “Pathwise” Product Rule

In traditional probability, the product rule for the joint distribution of random variables X and Y is written quite simply as $p(x, y) = p(x | y)p(y)$. For path measures defined on the space of paths of an Itô process, it is possible to write a similar rule, known as the *disintegration theorem*. First, let us define some notation.

Definition 3.20 (Product Spaces and Canonical Projection Operator). *Let $(\Omega_x, \mathcal{F}_x)$ and $(\Omega_y, \mathcal{F}_y)$ be two measurable spaces. In the product space $\Omega_x \times \Omega_y$, we define a measurable rectangle the set of the form $A_x \times A_y$, where $A_x \in \mathcal{F}_x$ and $A_y \in \mathcal{F}_y$. Then, the σ -algebra of $\Omega_x \times \Omega_y$ is defined as the σ -algebra generated by all measurable rectangles, resulting in the product space $(\Omega_x \times \Omega_y, \mathcal{F}_x \otimes \mathcal{F}_y)$. The canonical projection operators are measurable functions defined as*

$$\pi_x : \Omega \rightarrow \Omega_x, \quad \pi_x(\omega_x, \omega_y) = \omega_x, \quad \forall \omega_x \in \Omega_x, \omega_y \in \Omega_y \quad (3.43)$$

$$\pi_y : \Omega \rightarrow \Omega_y, \quad \pi_y(\omega_x, \omega_y) = \omega_y, \quad \forall \omega_x \in \Omega_x, \omega_y \in \Omega_y. \quad (3.44)$$

We can now state the disintegration theorem as follows:

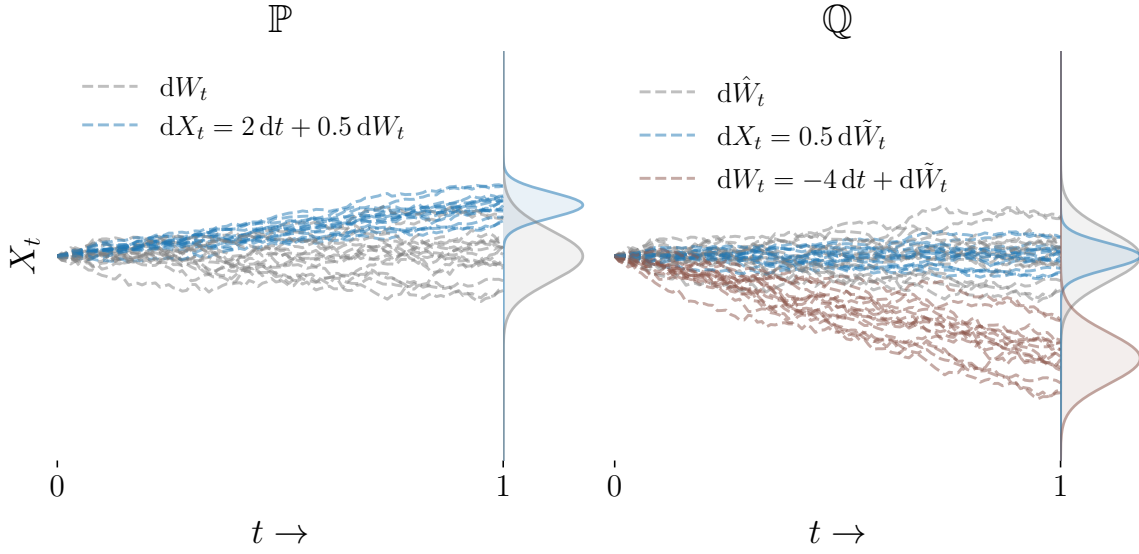


Fig. 3.2. An example of a change of measure using Girsanov's theorem. On the left, we plot the solution to an SDE $dX_t = a_t X_t dt + \sigma_t dW_t$, with $a_t = 1$ and $\sigma_t = 0.5$ along with the Brownian motion dW_t defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. On the right, we plot the same SDE under a change of measure, where we have a new Brownian motion $d\hat{W}_t$ defined on the probability space $(\Omega, \mathcal{F}, \mathbb{Q})$. Under this new measure, the process $d\hat{W}_t = dW_t + \frac{a_t}{\sigma_t} dt$ is a standard Brownian motion, and the process X_t can be written as $dX_t = \sigma_t d\hat{W}_t$. The original Brownian motion now has a negative drift.

Theorem 3.21 (Disintegration Theorem). *Let $(\Omega, \mathcal{B}(\Omega), \mathbb{P})$ be a product space where $\Omega = \Omega_x \times \Omega_y$, and a canonical projection operator π_x and π_y be defined as in Definition 3.20. Then, there exists a probability measure $\mathbb{P}_{y|x}(\cdot | x)$ on $(\Omega_x, \mathcal{F}_x)$ such that*

$$\int_{\Omega_x \times \Omega_y} f(x, y) d\mathbb{P}(x, y) = \int_{\Omega_x} \int_{\Omega_y} f(x, y) d\mathbb{P}_{y|x}(y | x) d\mathbb{P}_x(\pi_x^{-1}(x)), \quad (3.45)$$

where we denote $\mathbb{P}_x = \mathbb{P}(\pi_x^{-1}(x))$ as the marginal probability measure and $\mathbb{P}_{y|x}$ as the conditional probability measure.

3.4.2 Girsanov's Theorem: "Pathwise" Change of Measure

We have discussed the Radon-Nikodym derivative previously, as the ratio of two probability measures, and a function that allows us to express one probability measure in terms of another (usually the Lebesgue measure). Unfortunately, for path measures that are laws of SDEs, there doesn't exist a simple expression for the Radon-Nikodym derivative that maps between the path measure to the Lebesgue measure. However, we

are still able to represent the density of two path measures corresponding to two Itô processes, which allows us to perform a “pathwise” change of measure. As we will soon see, the ability to write down the density ratio between two path measures allows us to calculate expectations of functions of paths of an Itô process with respect to the law of another, much simpler, Itô process, which is a very useful property for importance sampling, and for representing solutions to SDEs in terms of simpler processes.

Here, we introduce Girsanov’s theorem, a groundbreaking result in the theory of stochastic processes, which allows us to represent the density of two path measures corresponding to two Itô processes, which allows us to perform a “pathwise” change of measure. Girsanov’s theorem is originally stated in a somewhat different form, which we discuss for completeness, before stating useful corollaries that are more suited for our discussion. Intuitively, Girsanov’s Theorem states that if we change the drift coefficient of an Itô process, the law of the process does not change dramatically, and the law of the new process is absolutely continuous with respect to the law of the original process, and can be computed explicitly in terms of their Radon-Nikodym derivative.

We first state the original theorem, which is a bit more general than what we will need, and then state a corollary that is more suited for our discussion.

Theorem 3.22 (Girsanov’s Theorem). *Let X_t be an Itô process defined on $(\Omega, \mathcal{F}, \mathbb{P})$ as the solution to the SDE*

$$dX_t = a(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 = x. \quad (3.46)$$

Suppose there exist $u(t, x)$ and $v(t, x)$ such that

$$u(t, x)\sigma(t, x) = a(t, x) - v(t, x), \quad \forall t \in [0, T], \quad x \in \mathbb{R}^n. \quad (3.47)$$

and $u(t, x)$ satisfies Novikov’s condition, given by

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T u(s, X_s)^2 ds \right) \right] < \infty. \quad (3.48)$$

Then, under the probability measure $\mathbb{Q}$ defined by the Radon-Nikodym derivative

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left(\int_0^T u(s, X_s) dW_s - \frac{1}{2} \int_0^T u(s, X_s)^2 ds \right), \quad (3.49)$$

the process

$$\hat{W}_t = W_t + \int_0^t u(s, X_s) ds \quad (3.50)$$

is a Wiener process under $\mathbb{Q}$, and in terms of the process $\hat{W}_t$, the process X_t can be written as

$$dX_t = v(t, X_t) dt + \sigma(t, X_t) d\hat{W}_t. \quad (3.51)$$

Proof. See Oksendal (2013, Theorem 8.6.6, Pg. 165). ■

As a concrete example of applying Theorem 3.22, we can see how an Itô process can be written as a Brownian motion under a change of measure.

Example 3.23 (Itô Process as Brownian Motion). Let X_t be an Itô process defined on $(\Omega, \mathcal{F}, \mathbb{P})$ as the solution to the SDE

$$dX_t = a_t(X_t) dt + \sigma_t dW_t, \quad X_0 = x. \quad (3.52)$$

Then, from Theorem 3.22, with a choice of $u(t, x) = \frac{a_t}{\sigma_t}$ and $v(t, x) = 0$, we have that $\hat{W}_t$ is a Brownian motion defined as

$$d\hat{W}_t = dW_t - \frac{a_t}{\sigma_t} dt, \quad \hat{W}_0 = 0, \quad (3.53)$$

under the probability measure $\mathbb{Q}$ which is defined through the Radon-Nikodym derivative

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left(\int_0^T \frac{a_s}{\sigma_s} dW_s - \frac{1}{2} \int_0^T \left(\frac{a_s}{\sigma_s} \right)^2 ds \right), \quad 0 \leq t \leq T, \quad (3.54)$$

and X_t can be written as a scaled Brownian motion under $\mathbb{Q}$ as

$$dX_t = \sigma_t d\hat{W}_t, \quad X_0 = x. \quad (3.55)$$

We can further state a more general version of Girsanov's theorem between general Itô processes that are solutions to SDEs as follows:

Theorem 3.24 (Girsanov's Theorem for RND Between SDEs). Given Itô processes X_t and Y_t defined as the solutions to the SDEs

$$dX_t = a_t(X_t) dt + \sigma_t dW_t, \quad X_0 \sim \mu, \quad (3.56)$$

and

$$dY_t = b_t(Y_t) dt + \nu_t dW_t, \quad Y_0 \sim \nu, \quad (3.57)$$

with path measures $\mathbb{P}^{a,\mu}$ and $\mathbb{P}^{b,\nu}$ respectively, then we have

$$\ln \left(\frac{d\mathbb{P}^{a,\mu}}{d\mathbb{P}^{b,\nu}} \right) (Z) = \ln \left(\frac{d\mu}{d\nu} \right) (Z_0) + \int_0^T \sigma_t^{-2} (a_t - b_t) (Z_t) \cdot dZ_t + \frac{1}{2} \int_0^T \sigma_t^{-2} (b_t^2 - a_t^2) (Z_t) dt. \quad (3.58)$$

With the ability to write down the density ratio between two path measures, we are now able to define well-known concepts from standard probability theory, but in the context of path measures. For example, we can define the KL divergence between two path measures as follows:

Definition 3.25 (KL Divergence Between Path Measures). *Let $\mathbb{P}^{a,\mu}$ and $\mathbb{P}^{b,\nu}$ be two path measures for Itô processes X_t and Y_t defined as the solutions to the SDEs with drift coefficients a_t and b_t respectively and diffusion coefficient σ_t . Then the KL divergence between $\mathbb{P}^{a,\mu}$ and $\mathbb{P}^{b,\nu}$ is defined as*

$$\mathcal{D}_{\text{KL}}(\mathbb{P}^{a,\mu} \parallel \mathbb{P}^{b,\nu}) = \mathbb{E}_{\mathbb{P}^{a,\mu}} \left[\ln \left(\frac{d\mu}{d\nu} \right) (X_0) + \frac{1}{2} \int_0^T \sigma_t^{-2} \|a_t - b_t\|^2 (X_t) dt \right]. \quad (3.59)$$

3.4.3 Doob's h -transform: SDEs with Boundary Conditions

Another useful property of Itô processes is the ability to condition an Itô process on a set of paths. This is particularly useful for solving SDEs with boundary conditions, which are SDEs that have constraints on the values of the process at the boundaries of the domain. Here, we discuss Doob's h -transform, a tool that allows us to define Itô processes conditioned on boundary conditions as the solution to an SDE with an additional term in the drift coefficient.

Definition 3.26 (Doob's h -transform). *Let X_t be an Itô process defined as the solution to the SDE*

$$dX_t = a_t(X_t) dt + \sigma_t dW_t, \quad X_0 = x. \quad (3.60)$$

It then follows that the conditioned process X_t conditioned on a boundary condition $X_T \in B$ is defined as the solution to the SDE

$$dX_t = \left(a_t(X_t) + \sigma_t^2 \nabla_{X_t} \ln \mathbb{P}(X_T \in B \mid X_t) \right) dt + \sigma_t dW_t, \quad X_0 = x, \quad (3.61)$$

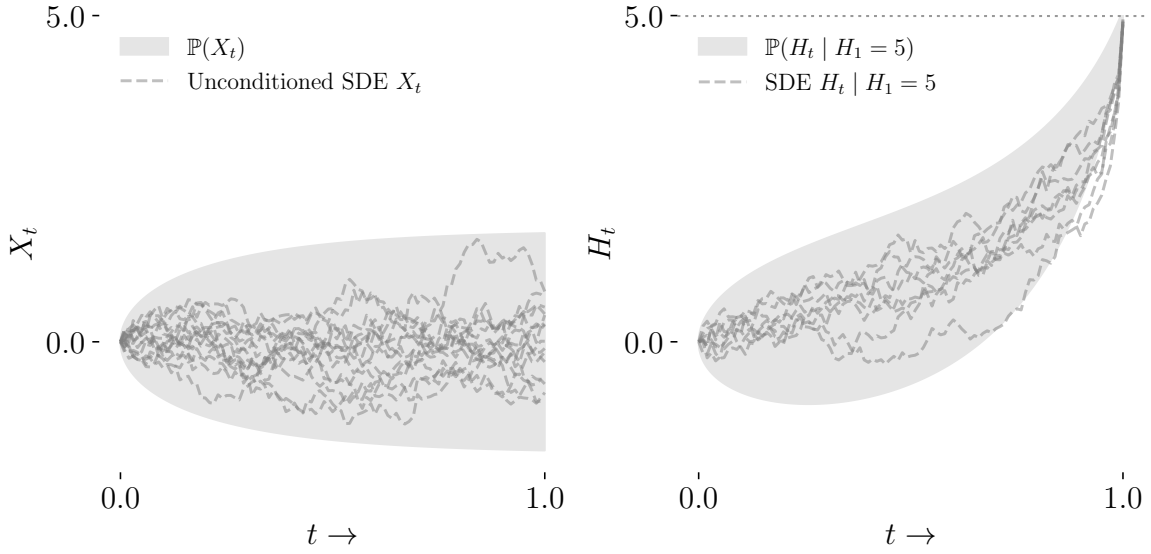


Fig. 3.3. An example of an unconditioned SDE (left) and an SDE conditioned on a boundary condition using Doob’s h -transform(right). On the left, we plot the solution to an SDE $dX_t = \mu_0 X_t dt + \sigma_0 dW_t$, with $\mu_0 = 1$ and $\sigma_0 = 0.5$. On the right, we plot the solution to the same SDE, but conditioned on the boundary condition $X_T = 5$. The conditioned SDE is given by $dX_t = (\mu_0 X_t + \sigma_0^2 \nabla_{X_t} \ln \mathbb{P}(X_T = 1 | X_t)) dt + \sigma_0 dW_t$.

where the term $h(X_t, t) \triangleq \mathbb{P}(X_T \in B | X_t)$ is defined as the h -transform, and we have $\mathbb{P}(X_T \in B) = 1$.

Proof. See Särkkä et al. (2019, Section 7.5, Pg. 113). ■

We have discussed some very useful properties of path measures, and techniques to perform “pathwise” operations on Itô processes, such as conditioning and a change of measure. We now turn our attention to one of the seminal results in stochastic calculus, which has enabled their application in generative modelling and diffusion models, known as the time-reversal of SDEs.

3.5 Time-Reversal of SDEs

As we alluded to in the previous chapter, generative modelling algorithms such as score-matching and DDPMs simultaneously design the “forward” process $q_\theta(z | x)$ that starts from an unknown data distribution $\hat{\pi}$ and iteratively noises it until a tractable noise distribution π_T , and the “backward” process $p_\phi(x | z)$ that starts from the noise distribution π_T and iteratively removes noise from it until the data distribution $\hat{\pi}$. An

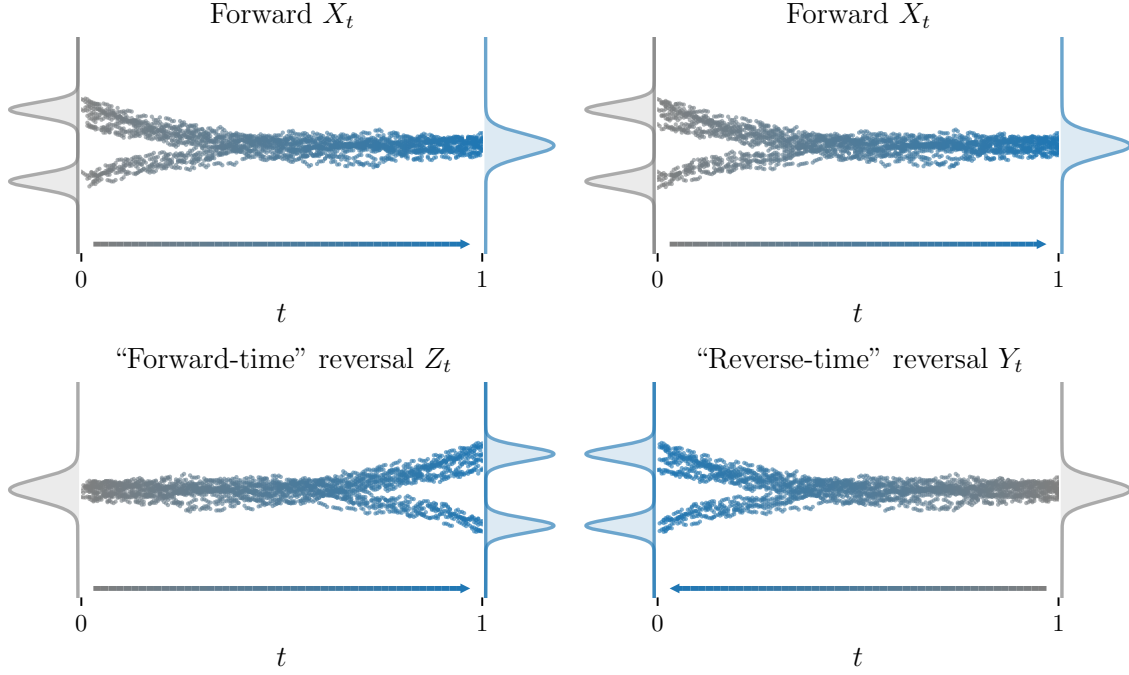


Fig. 3.4. Sample paths of a forward Ornstein–Uhlenbeck (OU) diffusion and its two distinct time-reversal conventions. The forward SDE is $dX_t = \theta(\mu - X_t) dt + \sigma dW_t$ on $t \in [0, 1]$ with $(\theta, \sigma, \mu) = (4, 1, 0)$ and bimodal start $X_0 \sim \frac{1}{2}\mathcal{N}(+2, 0.4^2) + \frac{1}{2}\mathcal{N}(-2, 0.4^2)$; the OU process converges to $\mathcal{N}(0, 1)$ by $t = 1$. Twenty paths are shown as dashed lines, where the grey $\rightarrow$ blue colour encodes the physical time that the process is solved in the direction of. The top row shows the forward process being solved from $t = 0$ to $t = 1$ in the forward direction (left to right). The bottom-left panel is the “forward-time” time reversal, where the re-indexed process $Z_t := X_{1-t}$ is another forward process adapted to the same filtration as X_t and is solved from $t = 0$ to $t = 1$ in the forward direction. The bottom right panel is the “reverse-time” time reversal, where the process $Y_t := X_t$ is defined backwards in time, and solved in the reverse direction from $t = 1$ to $t = 0$ against the reverse Brownian motion W_t . The solver runs right to left, and the color gradient is flipped accordingly.

alluring property of SDEs is that we can choose a specific form of the “forward” process q_θ as a simple SDE, and due to the time-reversal property of SDEs, we are able to write down the “backward” process $p\phi$ in closed form. Anderson and Nelson showed that it is possible to write down a “backward” Itô process that is the time-reversal of the forward process, so that the marginal distributions of the forward and backward processes are the same at the same time.

There are two distinct approaches on how to define the time-reversal SDE, both of which have subtle differences that can impact the theoretical properties of SDEs that

we can work with. The first approach, which we title as “forward-time”, defines the time-reversal SDE on the same filtration as the original forward process, with the same “forward-time” Brownian motion as the stochastic term. This approach allows us to use our discussed tools such as Girsanov’s theorem, and continue to use the definition of the Itô integral with respect to the forward-time Brownian motion.

The second approach, which we title as “reverse-time”; defines the time-reversal in a manner that might be more intuitive to consider from a theoretical perspective, but is more difficult to work with from a practical perspective due to the lack of current tooling. In this approach, we first define a “reverse-time” Brownian motion, which is a Brownian motion that integrates backwards in time, starting from the end-point of the original forward process. We then define the time-reversal SDE on a different filtration, which is the filtration of the reverse-time Brownian motion.

Definition 3.27 (Time-reversal of SDEs with forward time). *Given an Itô process X_t adapted to the “forward-time” filtration $\mathcal{F}_t = \sigma\{X_s, W_s : 0 \leq s \leq t\}$ which is defined as the solution to the SDE*

$$dX_t = \mu_t(X_t) dt + \sigma_t(X_t) dW_t, \quad X_0 \sim \hat{\pi}, \quad (3.62)$$

we introduce a re-indexed stochastic process $Z_t := X_{T-t}, t \in [0, T]$ adapted to the same “forward-time” filtration $\mathcal{F}_t$. Then, Z_t is the solution to the following SDE

$$dZ_t = \left[-\mu_{T-t}(Z_t) + \sigma_{T-t}^\top(Z_t) \nabla_{Z_t} \ln P_{T-t}(Z_t) \right] dt + \sigma_{T-t}(Z_t) dW_t, \quad Z_0 \sim \pi_T \quad (3.63)$$

There are some important properties of this time-reversal to note:

- By definition, Z_t is adapted to the same “forward-time” filtration $\mathcal{F}_t$ as the original process X_t . This means that we solve Z_t in the forward-time sense, with a Brownian motion W_t that is adapted to the same filtration $\mathcal{F}_t$.
- However, due to the re-indexing of the random variable Z_t , the joint law of the stochastic process *is not the same* as the joint law of the original process X_t . This is an important property to note, as the “forward-time” time-reversal matches the marginal distributions of X_t and Z_t , but not the joint law.

Definition 3.28 (Time-reversal of SDEs with reverse time). *Given an Itô process X_t adapted to the “forward-time” filtration $\mathcal{F}_t = \sigma\{X_s, W_s : 0 \leq s \leq t\}$ which*

is defined as the solution to the SDE

$$dX_t = \mu_t(X_t) dt + \sigma_t(X_t) dW_t, \quad X_0 \sim \hat{\pi}, \quad (3.64)$$

we then define the same random variable $Y_t := X_t, t \in [0, T]$ defined backwards in time, and adapted to the “reverse-time” filtration $\mathcal{G}_t = \sigma\{X_s, W_s : T \geq s \geq t\}$. Then, Y_t is the solution to the following SDE solved backwards in time from $t = T$ to $t = 0$

$$dY_t = \left[\mu_t(Y_t) - \sigma_t^\top \sigma_t \nabla_{Y_t} \ln P_t(Y_t) \right] dt + \sigma_t(Y_t) d\overleftarrow{W}_t, \quad Y_T \sim \pi_T, \quad (3.65)$$

where $\overleftarrow{W}_t$ is a Brownian motion that is adapted to the “reverse-time” filtration $\mathcal{G}_t$.

Some important properties of the “reverse-time” time-reversal to note:

- By definition, Y_t is adapted to the “reverse-time” filtration $\mathcal{G}_t$, and not the “forward-time” filtration $\mathcal{F}_t$. This means that we solve Y_t in the reverse-time sense, with a Brownian motion $\overleftarrow{W}_t$ that is adapted to the “reverse-time” filtration $\mathcal{G}_t$.
- Here, however, the joint law of the stochastic process is the same as the joint law of the original process X_t . This is an important property to note, as the “reverse-time” time-reversal matches the joint law of X_t and Y_t , i.e. $\text{Law}(\{X_t\}_{0 \leq t \leq T}) = \text{Law}(\{Y_t\}_{0 \leq t \leq T})$.

Theoretical Properties of Time-Reversals

Depending on the choice of time-reversal, there are certain mathematical properties from stochastic calculus that are preserved, and certain properties that are not preserved.

- Girsanov’s theorem: While we didn’t discuss the proof of Girsanov’s theorem in this chapter, an important property that is used in the proof is that the processes need to be adapted to the same filtration. Therefore, the “forward-time” time-reversal allows us to use Girsanov’s theorem between the forward and backward processes.
- Mixing an Itô integral that is forward-adapted (to $\mathcal{F}_t$) with an Itô integral that is reverse-adapted (to $\mathcal{G}_t$) breaks Burkholder–Davis–Gundy inequalities and Itô isometry. Hence expectations and L^2 bounds that are routine in the forward-only world become really difficult to work with in the “reverse-time” world.

- However, in our framework for generative modelling and sampling, we are really interested in learning our parametrised maps p_ϕ and q_θ under the framework of probability divergences that we want to write between two path measures $\mathbb{P}$ and $\mathbb{Q}$, corresponding to the forward and backward processes respectively. In this situation, the “reverse-time” time-reversal allows us to write the KL divergence between the forward and backward processes easily, as we will readily minimise the KL divergence between the forward and “reverse-time” backward processes when we use the “reverse-time” time-reversal, since their joint law is the same.

3.6 Reverse-Time SDEs and Föllmer’s Itô Calculus

We have seen that the “reverse-time” SDE framework provides the same joint law for the forward and reverse SDEs, which would enable us to define probability divergences between the forward and reverse SDEs. However, since the forward and reverse SDEs exist on different filtrations going in different directions, we cannot rely on our existing machinery of Ito integrals and Girsanov’s theorem, bringing us to a standstill.

In this section, we introduce an alternative framework, Föllmer’s Itô calculus, that allows us to derive a version of Itô’s formula for SDEs without relying on the existence of filtrations. While there are some subtleties to this framework, and the results might be not as general as the existing framework, for our purposes, it suffices to use this framework to define a novel Radon-Nikodym derivative between the forward and reverse processes, which we will use to define a novel Girsanov’s theorem in the next chapter.

Definition 3.29 (Itô’s formula without probability). (*Eberle (2011, Theorem 6.6, Pg. 122), Föllmer (2006)*) Suppose that $X : [0, T) \rightarrow \mathbb{R}^d$ is a continuous function with continuous quadratic variation $\langle X \rangle_t$ w.r.t a partition (π_n) . Then, for any $f \in C^2([0, t] \times \mathbb{R}^d)$, and for any $t \in [0, T)$, the Itô integral

$$\int_0^t \nabla f(s, X_s) dX_s := \lim_{n \rightarrow \infty} \sum_{s \in \pi_n, s < t} \nabla f(s, X_s) (X_{s' \wedge t} - X_s) \quad (3.66)$$

exists, and Itô’s formula

$$f(X_t) = f(X_0) + \int_0^t \nabla f(s, X_s) dX_s + \frac{1}{2} \nabla^2 f(s, X_s) d\langle X \rangle_s \quad (3.67)$$

holds.

We note here that Definition 3.29 holds only for individual continuous functions (or paths) X_t with continuous quadratic variation, and not for general processes adapted to a filtration. Therefore, we no longer require the existence of a filtration to define Itô's formula. However, it is important to note that the result is dependent on the specific partition of the interval $[0, t)$ that we choose, and if the quadratic variation of the function X_t varies with the partition, then the result may not hold. This is not a problem for our purposes, as the quadratic variation of the Wiener process is deterministic, finite and continuous regardless of the partition almost surely (Eberle (2011, Pg. 120)).

In Föllmer's framework, defining the "reverse-time" Itô integral is done by using the same definition of the Itô integral as in the forward-time case.

Definition 3.30 ("Reverse-time" Itô's formula without probability). (Eberle (2011, Theorem 6.7, Pg. 123), Föllmer (2006) and Kunita (1982)) Suppose that $X : [0, T) \rightarrow \mathbb{R}^d$ is a continuous function with continuous quadratic variation $\langle X \rangle_t$ w.r.t a partition (π_n) . Then, for any $f \in C^2([0, t) \times \mathbb{R}^d)$, and for any $t \in [0, T)$, the backward Itô integral

$$\int_0^t \nabla f(s, X_s) d\overleftarrow{X}_s := \lim_{n \rightarrow \infty} \sum_{s \in \pi_n, s < t} \nabla f(s' \wedge t, X_s)(X_{s' \wedge t} - X_s) \quad (3.68)$$

exists, and Itô's formula

$$f(X_t) = f(X_0) + \int_0^t \nabla f(s, X_s) d\overleftarrow{X}_s - \frac{1}{2} \nabla^2 f(s, X_s) d\langle X \rangle_s \quad (3.69)$$

holds.

Chapter 4

Continuous-Time Markov Chains and Diffusion Processes

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Appendix A

How to install L^AT_EX

Windows OS

TeXLive package - full version

1. Download the TeXLive ISO (2.2GB) from <https://www.tug.org/texlive/>
2. Download WinCDEmu (if you don't have a virtual drive) from <http://wincdemu.sysprogs.org/download/>
3. To install Windows CD Emulator follow the instructions at <http://wincdemu.sysprogs.org/tutorials/install/>
4. Right click the iso and mount it using the WinCDEmu as shown in <http://wincdemu.sysprogs.org/tutorials/mount/>
5. Open your virtual drive and run setup.pl

or

Basic MikTeX - T_EX distribution

1. Download Basic-MiK_TE_X(32bit or 64bit) from <http://miktex.org/download>
2. Run the installer

3. To add a new package go to Start » All Programs » MikTeX » Maintenance (Admin) and choose Package Manager
4. Select or search for packages to install

TexStudio - T_EX editor

1. Download TexStudio from
<http://texstudio.sourceforge.net/#downloads>
2. Run the installer

Mac OS X

MacTeX - T_EX distribution

1. Download the file from
<https://www.tug.org/mactex/>
2. Extract and double click to run the installer. It does the entire configuration, sit back and relax.

TexStudio - T_EX editor

1. Download TexStudio from
<http://texstudio.sourceforge.net/#downloads>
2. Extract and Start

Unix/Linux

TeXLive - T_EX distribution

Getting the distribution:

1. TeXLive can be downloaded from
<http://www.tug.org/texlive/acquire-netinstall.html>.
2. TeXLive is provided by most operating system you can use (rpm,apt-get or yum) to get TeXLive distributions

Installation

1. Mount the ISO file in the mnt directory

```
mount -t iso9660 -o ro,loop,noauto /your/texlive####.iso /mnt
```

2. Install wget on your OS (use rpm, apt-get or yum install)
3. Run the installer script install-tl.

```
cd /your/download/directory
./install-tl
```

4. Enter command 'i' for installation

5. Post-Installation configuration:

<http://www.tug.org/texlive/doc/texlive-en/texlive-en.html#x1-320003.4.1>

6. Set the path for the directory of TexLive binaries in your .bashrc file

For 32bit OS

For Bourne-compatible shells such as bash, and using Intel x86 GNU/Linux and a default directory setup as an example, the file to edit might be

```
edit ~/.bashrc file and add following lines
PATH=/usr/local/texlive/2011/bin/i386-linux:$PATH;
export PATH
MANPATH=/usr/local/texlive/2011/texmf/doc/man:$MANPATH;
export MANPATH
INFOPATH=/usr/local/texlive/2011/texmf/doc/info:$INFOPATH;
export INFOPATH
```

For 64bit OS

```
edit ~/.bashrc file and add following lines
PATH=/usr/local/texlive/2011/bin/x86_64-linux:$PATH;
export PATH
MANPATH=/usr/local/texlive/2011/texmf/doc/man:$MANPATH;
export MANPATH
INFOPATH=/usr/local/texlive/2011/texmf/doc/info:$INFOPATH;
export INFOPATH
```

Fedora/RedHat/CentOS:

```
sudo yum install texlive
sudo yum install psutils
```

SUSE:

```
sudo zypper install texlive
```

Debian/Ubuntu:

```
sudo apt-get install texlive texlive-latex-extra
sudo apt-get install psutils
```


Appendix B

Installing the CUED class file

$\LaTeX$.cls files can be accessed system-wide when they are placed in the $\langle\text{texmf}\rangle/\text{tex}/\text{latex}$ directory, where $\langle\text{texmf}\rangle$ is the root directory of the user's $\text{T}_{\text{E}}\text{X}$ installation. On systems that have a local texmf tree ($\langle\text{texmflocal}\rangle$), which may be named “texmf-local” or “localtexmf”, it may be advisable to install packages in $\langle\text{texmflocal}\rangle$, rather than $\langle\text{texmf}\rangle$ as the contents of the former, unlike that of the latter, are preserved after the $\LaTeX$ system is reinstalled and/or upgraded.

It is recommended that the user create a subdirectory $\langle\text{texmf}\rangle/\text{tex}/\text{latex}/\text{CUED}$ for all CUED related $\LaTeX$ class and package files. On some $\LaTeX$ systems, the directory look-up tables will need to be refreshed after making additions or deletions to the system files. For $\text{T}_{\text{E}}\text{X}$ Live systems this is accomplished via executing “texhash” as root. $\text{M}_{\text{I}}\text{K}_{\text{T}}\text{E}_{\text{X}}$ users can run “initexmf -u” to accomplish the same thing.

Users not willing or able to install the files system-wide can install them in their personal directories, but will then have to provide the path (full or relative) in addition to the filename when referring to them in $\LaTeX$.

